Tab 1

MICROFAB.txt :


```
{
  "port": 9090,
  "endorsing_organizations": [
    {
      "name": "StartupOrg",
      "peers": [
        {
          "name": "peer0",
          "stateDatabase": "couchdb"
        }
      ]
    },
    {
      "name": "ValidatorOrg",
      "peers": [
        {
          "name": "peer0",
          "stateDatabase": "couchdb"
        }
      ]
    },
    {
      "name": "InvestorOrg",
      "peers": [
        {
          "name": "peer0",
          "stateDatabase": "couchdb"
        }
      ]
    },
    {
      "name": "PlatformOrg",
      "peers": [
        {
          "name": "peer0",
          "stateDatabase": "couchdb"
        }
      ]
    }
  ],

  "channels": [
    {
      "name": "gov-validation-channel",
      "endorsing_organizations": [
        "StartupOrg",
        "ValidatorOrg",
```

```
      "PlatformOrg"
    ]
  },
  {
    "name": "investment-execution-channel",
    "endorsing_organizations": [
      "StartupOrg",
      "ValidatorOrg",
      "InvestorOrg",
      "PlatformOrg"
    ]
  }
 ]
}
```

Commands right after this to run :

1. export MICROFAB_CONFIG=$(cat MICROFAB.txt)
2. docker run --name microfab -e MICROFAB_CONFIG -p 9090:9090 ibmcom/ibp-microfab

Right after that generate wallets msp gateways

```
curl -s http://console.127-0-0-1.nip.io:9090/ak/api/v1/components | weft microfab -w ./_wallets -p ./_gateways -m ./_msp -f
```

Terminal reference right after this command you get env variables to set make one new terminal for every org and set env acc not the ca normal orgmsp

```
aditidubey@Kedarnath:/mnt/c/Users/aditi/Downloads/crowdfunding_cc_2$ curl -s http://console.127-0-0-1.nip.io:9090/ak/api/v1/components | weft microfab -w ./_wallets -p ./_gateways -m ./_msp -f
Gateway profile written to : /mnt/c/Users/aditi/Downloads/crowdfunding_cc_2/_gateways/investororggateway.json
Gateway profile written to : /mnt/c/Users/aditi/Downloads/crowdfunding_cc_2/_gateways/platformorggateway.json
Gateway profile written to : /mnt/c/Users/aditi/Downloads/crowdfunding_cc_2/_gateways/startuporggateway.json
Gateway profile written to : /mnt/c/Users/aditi/Downloads/crowdfunding_cc_2/_gateways/validatororggateway.json
Added identity under label investororgcaadmin to the wallet at /mnt/c/Users/aditi/Downloads/crowdfunding_cc_2/_wallets/InvestorOrg
Added identity under label platformorgadmin to the wallet at /mnt/c/Users/aditi/Downloads/crowdfunding_cc_2/_wallets/PlatformOrg
Added identity under label startuporgcaadmin to the wallet at /mnt/c/Users/aditi/Downloads/crowdfunding_cc_2/_wallets/StartupOrg
Added identity under label validatororgadmin to the wallet at /mnt/c/Users/aditi/Downloads/crowdfunding_cc_2/_wallets/ValidatorOrg
Added identity under label validatororgcaadmin to the wallet at /mnt/c/Users/aditi/Downloads/crowdfunding_cc_2/_wallets/ValidatorOrg
```

Added identity under label ordereradmin to the wallet at
/mnt/c/Users/aditi/Downloads/crowdfunding_cc_2/_wallets/Orderer
Added identity under label investororgadmin to the wallet at
/mnt/c/Users/aditi/Downloads/crowdfunding_cc_2/_wallets/InvestorOrg
Added identity under label platformorgcaadmin to the wallet at
/mnt/c/Users/aditi/Downloads/crowdfunding_cc_2/_wallets/PlatformOrg
Added identity under label startuporgadmin to the wallet at
/mnt/c/Users/aditi/Downloads/crowdfunding_cc_2/_wallets/StartupOrg

Environment variables:

For investororgcaadmin @ InvestorOrg use these:

export CORE_PEER_LOCALMSPID=InvestorOrgMSP
export
CORE_PEER_MSPCONFIGPATH=/mnt/c/Users/aditi/Downloads/crowdfunding_cc_2/_msp/InvestorOrg/investororgcaadmin/msp
export CORE_PEER_ADDRESS=investororgpeer-api.127-0-0-1.nip.io:9090

For investororgadmin @ InvestorOrg use these:

export CORE_PEER_LOCALMSPID=InvestorOrgMSP
export
CORE_PEER_MSPCONFIGPATH=/mnt/c/Users/aditi/Downloads/crowdfunding_cc_2/_msp/InvestorOrg/investororgadmin/msp
export CORE_PEER_ADDRESS=investororgpeer-api.127-0-0-1.nip.io:9090

For platformorgadmin @ PlatformOrg use these:

export CORE_PEER_LOCALMSPID=PlatformOrgMSP
export
CORE_PEER_MSPCONFIGPATH=/mnt/c/Users/aditi/Downloads/crowdfunding_cc_2/_msp/PlatformOrg/platformorgadmin/msp
export CORE_PEER_ADDRESS=platformorgpeer-api.127-0-0-1.nip.io:9090

For platformorgcaadmin @ PlatformOrg use these:

export CORE_PEER_LOCALMSPID=PlatformOrgMSP
export
CORE_PEER_MSPCONFIGPATH=/mnt/c/Users/aditi/Downloads/crowdfunding_cc_2/_msp/PlatformOrg/platformorgcaadmin/msp
export CORE_PEER_ADDRESS=platformorgpeer-api.127-0-0-1.nip.io:9090

For startuporgcaadmin @ StartupOrg use these:

export CORE_PEER_LOCALMSPID=StartupOrgMSP

```
export
CORE_PEER_MSPCONFIGPATH=/mnt/c/Users/aditi/Downloads/crowdfunding_cc_2/_msp/
StartupOrg/startuporgcaadmin/msp
export CORE_PEER_ADDRESS=startuporgpeer-api.127-0-0-1.nip.io:9090

For startuporgadmin @  StartupOrg use these:

export CORE_PEER_LOCALMSPID=StartupOrgMSP
export
CORE_PEER_MSPCONFIGPATH=/mnt/c/Users/aditi/Downloads/crowdfunding_cc_2/_msp/
StartupOrg/startuporgadmin/msp
export CORE_PEER_ADDRESS=startuporgpeer-api.127-0-0-1.nip.io:9090

For validatororgadmin @  ValidatorOrg use these:

export CORE_PEER_LOCALMSPID=ValidatorOrgMSP
export
CORE_PEER_MSPCONFIGPATH=/mnt/c/Users/aditi/Downloads/crowdfunding_cc_2/_msp/
ValidatorOrg/validatororgadmin/msp
export CORE_PEER_ADDRESS=validatororgpeer-api.127-0-0-1.nip.io:9090

For validatororgcaadmin @  ValidatorOrg use these:

export CORE_PEER_LOCALMSPID=ValidatorOrgMSP
export
CORE_PEER_MSPCONFIGPATH=/mnt/c/Users/aditi/Downloads/crowdfunding_cc_2/_msp/
ValidatorOrg/validatororgcaadmin/msp
export CORE_PEER_ADDRESS=validatororgpeer-api.127-0-0-1.nip.io:9090
aditidubey@Kedarnath:/mnt/c/Users/aditi/Downloads/crowdfunding_cc_2$ Installing
Binaries: curl -sSL
https://raw.githubusercontent.com/hyperledger/fabric/main/scripts/install-fabric.sh | bash -s --
binary
Installing: command not found
aditidubey@Kedarnath:/mnt/c/Users/aditi/Downloads/crowdfunding_cc_2$ curl -sSL
https://raw.githubusercontent.com/hyperledger/fabric/main
/scripts/install-fabric.sh | bash -s -- binary

Pull Hyperledger Fabric binaries

===> Downloading version 2.5.14 platform specific fabric binaries
===> Downloading:
https://github.com/hyperledger/fabric/releases/download/v2.5.14/hyperledger-fabric-linux-am
d64-2.5.14.tar.gz
===> Will unpack to: /mnt/c/Users/aditi/Downloads/crowdfunding_cc_2
  % Total    % Received % Xferd  Average Speed   Time    Time     Time  Current
                                 Dload  Upload   Total   Spent    Left  Speed
  0     0    0     0    0     0      0      0 --:--:-- --:--:-- --:--:--     0
100  119M  100  119M    0     0   919k      0  0:02:13  0:02:13 --:--:--  905k
```

```
==> Done.
===> Downloading version 1.5.15 platform specific fabric-ca-client binary
===> Downloading:
https://github.com/hyperledger/fabric-ca/releases/download/v1.5.15/hyperledger-fabric-ca-lin
ux-amd64-1.5.15.tar.gz
===> Will unpack to: /mnt/c/Users/aditi/Downloads/crowdfunding_cc_2
  % Total    % Received % Xferd  Average Speed   Time    Time     Time  Current
                                 Dload  Upload   Total   Spent    Left  Speed
  0     0    0     0    0     0      0      0 --:--:-- --:--:-- --:--:--     0
100 30.0M  100 30.0M    0     0   858k      0  0:00:35  0:00:35 --:--:--  754k
==> Done.
aditidubey@Kedarnath:/mnt/c/Users/aditi/Downloads/crowdfunding_cc_2$ peer version
Command 'peer' not found, did you mean:
  command 'peet' from deb pipexec (2.6.1-3)
  command 'pee' from deb moreutils (0.67-1)
  command 'seer' from deb seer (1.1.4-7build2)
  command 'pear' from deb php-pear (1:1.10.13+submodules+notgz+2022032202-2build1)
  command 'peek' from deb peek (1.5.1+git20230114-1)
Try: sudo apt install <deb name>
aditidubey@Kedarnath:/mnt/c/Users/aditi/Downloads/crowdfunding_cc_2$  export
PATH=$PATH:${PWD}/bin
    export FABRIC_CFG_PATH=${PWD}/config
aditidubey@Kedarnath:/mnt/c/Users/aditi/Downloads/crowdfunding_cc_2$ peer version
peer:
 Version: v2.5.14
 Commit SHA: 24767ba
 Go version: go1.25.2
 OS/Arch: linux/amd64
 Chaincode:
  Base Docker Label: org.hyperledger.fabric
  Docker Namespace: hyperledger
```

Installing Binaries:

```
curl -sSL https://raw.githubusercontent.com/hyperledger/fabric/main/scripts/install-fabric.sh |
bash -s -- binary
```

Right after that set fabric env

```
export PATH=$PATH:${PWD}/bin
export FABRIC_CFG_PATH=${PWD}/config
```

Startup terminal

```
aditidubey@Kedarnath:/mnt/c/Users/aditi/Downloads/crowdfunding_cc_2$ export
CORE_PEER_LOCALMSPID=StartupOrgMSP
```

```
export
CORE_PEER_MSPCONFIGPATH=/mnt/c/Users/aditi/Downloads/crowdfunding_cc_2/_msp/
StartupOrg/startuporgadmin/msp
export CORE_PEER_ADDRESS=startuporgpeer-api.127-0-0-1.nip.io:9090
aditidubey@Kedarnath:/mnt/c/Users/aditi/Downloads/crowdfunding_cc_2$  export
PATH=$PATH:${PWD}/bin
    export FABRIC_CFG_PATH=${PWD}/config
aditidubey@Kedarnath:/mnt/c/Users/aditi/Downloads/crowdfunding_cc_2$ peer version
peer:
 Version: v2.5.14


 Commit SHA: 24767ba
 Go version: go1.25.2
 OS/Arch: linux/amd64
 Chaincode:
  Base Docker Label: org.hyperledger.fabric
  Docker Namespace: hyperledger

aditidubey@Kedarnath:/mnt/c/Users/aditi/Downloads/crowdfunding_cc_2$ peer lifecycle
chaincode install governancecc.tar.gz
2026-02-07 14:59:36.066 UTC 0001 INFO [cli.lifecycle.chaincode] submitInstallProposal ->
Installed remotely: response:<status:200
payload:"\nOgovernancecc_1:8e7e61cdd46e166483b9282a4021b18a6774eea3d55cf8849b
f8656e58bbf574\022\016governancecc_1" >
2026-02-07 14:59:36.067 UTC 0002 INFO [cli.lifecycle.chaincode] submitInstallProposal ->
Chaincode code package identifier:
governancecc_1:8e7e61cdd46e166483b9282a4021b18a6774eea3d55cf8849bf8656e58bbf
574
aditidubey@Kedarnath:/mnt/c/Users/aditi/Downloads/crowdfunding_cc_2$ peer lifecycle
chaincode install investmentcc_v2.tar.gz
2026-02-07 15:00:00.968 UTC 0001 INFO [cli.lifecycle.chaincode] submitInstallProposal ->
Installed remotely: response:<status:200
payload:"\nOinvestmentcc_1:0b366fdfd184bc98f2cee449c7f7631ca4880d5d598b6c532d72
a142c143aa44\022\016investmentcc_1" >
2026-02-07 15:00:00.969 UTC 0002 INFO [cli.lifecycle.chaincode] submitInstallProposal ->
Chaincode code package identifier:
investmentcc_1:0b366fdfd184bc98f2cee449c7f7631ca4880d5d598b6c532d72a142c143aa4
4
aditidubey@Kedarnath:/mnt/c/Users/aditi/Downloads/crowdfunding_cc_2$ peer lifecycle
chaincode approveformyorg \
  --channelID gov-validation-channel \
  --name governancecc \
  --version 1.0 \
  --sequence 1 \
  --package-id
governancecc_1:8e7e61cdd46e166483b9282a4021b18a6774eea3d55cf8849bf8656e58bbf
574
```

```
2026-02-07 15:03:52.318 UTC 0001 INFO [cli.lifecycle.chaincode] setOrdererClient ->
Retrieved channel (gov-validation-channel) orderer endpoint:
orderer-api.127-0-0-1.nip.io:9090
2026-02-07 15:03:53.315 UTC 0002 INFO [chaincodeCmd] ClientWait -> txid
[7e7ba1f1d89ca4f16c440d0b911ce99682186fce39d7546f6e6f04b232765236] committed
with status (VALID) at startuporgpeer-api.127-0-0-1.nip.io:9090
  --version 1.0 \
  --sequence 1 \
  --package-id
governancecc_1:8e7e61cdd46e166483b9282a4021b18a6774eea3d55cf8849bf8656e58bbf
574
2026-02-07 15:03:52.318 UTC 0001 INFO [cli.lifecycle.chaincode] setOrdererClient ->
Retrieved channel (gov-validation-channel) orderer endpoint:
orderer-api.127-0-0-1.nip.io:9090
2026-02-07 15:03:53.315 UTC 0002 INFO [chaincodeCmd] ClientWait -> txid
[7e7ba1f1d89ca4f16c440d0b911ce99682186fce39d7546f6e6f04b232765236] committed
with status (VALID) at startuporgpeer-api.127-0-0-1.nip.io:9090
ndpoint: orderer-api.127-0-0-1.nip.io:9090
2026-02-07 15:03:53.315 UTC 0002 INFO [chaincodeCmd] ClientWait -> txid
[7e7ba1f1d89ca4f16c440d0b911ce99682186fce39d7546f6e6f04b232765236] committed
with status (VALID) at startuporgpeer-api.127-0-0-1.nip.io:9090
2026-02-07 15:03:53.315 UTC 0002 INFO [chaincodeCmd] ClientWait -> txid
[7e7ba1f1d89ca4f16c440d0b911ce99682186fce39d7546f6e6f04b232765236] committed
with status (VALID) at startuporgpeer-api.127-0-0-1.nip.io:9090
6] committed with status (VALID) at startuporgpeer-api.127-0-0-1.nip.io:9090
aditidubey@Kedarnath:/mnt/c/Users/aditi/Downloads/crowdfunding_cc_2$ peer lifecycle
chaincode approveformyorg \
  --channelID investment-execution-channel \
  --name investmentcc \
  --version 1.0 \
  --sequence 1 \
  --package-id
investmentcc_1:0b366fdfd184bc98f2cee449c7f7631ca4880d5d598b6c532d72a142c143aa4
4
2026-02-07 15:05:03.194 UTC 0001 INFO [cli.lifecycle.chaincode] setOrdererClient ->
Retrieved channel (investment-execution-channel) orderer endpoint:
orderer-api.127-0-0-1.nip.io:9090
2026-02-07 15:05:03.514 UTC 0002 INFO [chaincodeCmd] ClientWait -> txid
[bd483ce16c74ce841c2a995cae126372af888a567e42e133bdcce336c1844f1b] committed
with status (VALID) at startuporgpeer-api.127-0-0-1.nip.io:9090
aditidubey@Kedarnath:/mnt/c/Users/aditi/Downloads/crowdfunding_cc_2$
```

Validator terminal

```
aditidubey@Kedarnath:/mnt/c/Users/aditi/Downloads/crowdfunding_cc_2$ export
CORE_PEER_LOCALMSPID=ValidatorOrgMSP
```

```
export
CORE_PEER_MSPCONFIGPATH=/mnt/c/Users/aditi/Downloads/crowdfunding_cc_2/_msp/
ValidatorOrg/validatororgadmin/msp
export CORE_PEER_ADDRESS=validatororgpeer-api.127-0-0-1.nip.io:9090
aditidubey@Kedarnath:/mnt/c/Users/aditi/Downloads/crowdfunding_cc_2$  export
PATH=$PATH:${PWD}/bin
    export FABRIC_CFG_PATH=${PWD}/config
aditidubey@Kedarnath:/mnt/c/Users/aditi/Downloads/crowdfunding_cc_2$ peer lifecycle
chaincode install governancecc.tar.gz
Error: chaincode install failed with status: 500 - failed to invoke backing implementation of
'InstallChaincode': could not build chaincode: external builder failed: external builder failed to
build: external builder 'golang' failed: exit status 1
aditidubey@Kedarnath:/mnt/c/Users/aditi/Downloads/crowdfunding_cc_2$ cd contracts
aditidubey@Kedarnath:/mnt/c/Users/aditi/Downloads/crowdfunding_cc_2/contracts$ go mod
tidy
aditidubey@Kedarnath:/mnt/c/Users/aditi/Downloads/crowdfunding_cc_2/contracts$ go build
# crowdfundingcc
./investmentcc.go:169:6: main redeclared in this block
        ./governancecc.go:179:6: other declaration of main
aditidubey@Kedarnath:/mnt/c/Users/aditi/Downloads/crowdfunding_cc_2/contracts$ cd ..
aditidubey@Kedarnath:/mnt/c/Users/aditi/Downloads/crowdfunding_cc_2$ peer lifecycle
chaincode install governancecc.tar.gz
Error: chaincode install failed with status: 500 - failed to invoke backing implementation of
'InstallChaincode': could not build chaincode: external builder failed: external builder failed to
build: external builder 'golang' failed: exit status 1
aditidubey@Kedarnath:/mnt/c/Users/aditi/Downloads/crowdfunding_cc_2$ peer lifecycle
chaincode package governancecc.tar.gz \
  --path ./contracts \
  --lang golang \
  --label governancecc_1
aditidubey@Kedarnath:/mnt/c/Users/aditi/Downloads/crowdfunding_cc_2$ peer lifecycle
chaincode install governancecc.tar.gz
Error: chaincode install failed with status: 500 - failed to invoke backing implementation of
'InstallChaincode': could not build chaincode: external builder failed: external builder failed to
build: external builder 'golang' failed: exit status 2
aditidubey@Kedarnath:/mnt/c/Users/aditi/Downloads/crowdfunding_cc_2$ cd contracts
aditidubey@Kedarnath:/mnt/c/Users/aditi/Downloads/crowdfunding_cc_2/contracts$ go mod
tidy
go: errors parsing go.mod:
go.mod:3: go directive expects exactly one argument
aditidubey@Kedarnath:/mnt/c/Users/aditi/Downloads/crowdfunding_cc_2/contracts$ go mod
tidy
aditidubey@Kedarnath:/mnt/c/Users/aditi/Downloads/crowdfunding_cc_2/contracts$ cd ..
aditidubey@Kedarnath:/mnt/c/Users/aditi/Downloads/crowdfunding_cc_2$ peer lifecycle
chaincode install governancecc.tar.gz
Error: chaincode install failed with status: 500 - failed to invoke backing implementation of
'InstallChaincode': could not build chaincode: external builder failed: external builder failed to
build: external builder 'golang' failed: exit status 2
```

```
aditidubey@Kedarnath:/mnt/c/Users/aditi/Downloads/crowdfunding_cc_2$ go mod tidy
go: go.mod file not found in current directory or any parent directory; see 'go help modules'
aditidubey@Kedarnath:/mnt/c/Users/aditi/Downloads/crowdfunding_cc_2$ cd contracts
aditidubey@Kedarnath:/mnt/c/Users/aditi/Downloads/crowdfunding_cc_2/contracts$ go mod
tidy
aditidubey@Kedarnath:/mnt/c/Users/aditi/Downloads/crowdfunding_cc_2/contracts$ cd ..
aditidubey@Kedarnath:/mnt/c/Users/aditi/Downloads/crowdfunding_cc_2$ peer lifecycle
chaincode install governancecc.tar.gz
Error: chaincode install failed with status: 500 - failed to invoke backing implementation of
'InstallChaincode': could not build chaincode: external builder failed: external builder failed to
build: external builder 'golang' failed: exit status 2
aditidubey@Kedarnath:/mnt/c/Users/aditi/Downloads/crowdfunding_cc_2$ peer lifecycle
chaincode install governancecc.tar.gz
Error: chaincode install failed with status: 500 - failed to invoke backing implementation of
'InstallChaincode': could not build chaincode: external builder failed: external builder failed to
build: external builder 'golang' failed: exit status 1
aditidubey@Kedarnath:/mnt/c/Users/aditi/Downloads/crowdfunding_cc_2$ cd contracts
aditidubey@Kedarnath:/mnt/c/Users/aditi/Downloads/crowdfunding_cc_2/contracts$ cd
governancecc
aditidubey@Kedarnath:/mnt/c/Users/aditi/Downloads/crowdfunding_cc_2/contracts/governa
ncecc$ go mod tidy
aditidubey@Kedarnath:/mnt/c/Users/aditi/Downloads/crowdfunding_cc_2/contracts/governa
ncecc$ go build
aditidubey@Kedarnath:/mnt/c/Users/aditi/Downloads/crowdfunding_cc_2/contracts/governa
ncecc$ cd ..
aditidubey@Kedarnath:/mnt/c/Users/aditi/Downloads/crowdfunding_cc_2/contracts$ cd ..
aditidubey@Kedarnath:/mnt/c/Users/aditi/Downloads/crowdfunding_cc_2$ peer lifecycle
chaincode package governancecc.tar.gz \
  --path ./contracts/governancecc \
  --lang golang \
  --label governancecc_1
aditidubey@Kedarnath:/mnt/c/Users/aditi/Downloads/crowdfunding_cc_2$ peer lifecycle
chaincode install governancecc.tar.gz
2026-02-07 14:56:42.777 UTC 0001 INFO [cli.lifecycle.chaincode] submitInstallProposal ->
Installed remotely: response:<status:200
payload:"\nOgovernancecc_1:8e7e61cdd46e166483b9282a4021b18a6774eea3d55cf8849b
f8656e58bbf574\022\016governancecc_1" >
2026-02-07 14:56:42.778 UTC 0002 INFO [cli.lifecycle.chaincode] submitInstallProposal ->
Chaincode code package identifier:
governancecc_1:8e7e61cdd46e166483b9282a4021b18a6774eea3d55cf8849bf8656e58bbf
574
aditidubey@Kedarnath:/mnt/c/Users/aditi/Downloads/crowdfunding_cc_2$ cd contracts
aditidubey@Kedarnath:/mnt/c/Users/aditi/Downloads/crowdfunding_cc_2/contracts$ cd
investmentcc
aditidubey@Kedarnath:/mnt/c/Users/aditi/Downloads/crowdfunding_cc_2/contracts/investme
ntcc$ go mod tidy
aditidubey@Kedarnath:/mnt/c/Users/aditi/Downloads/crowdfunding_cc_2/contracts/investme
ntcc$ go build
```

```
aditidubey@Kedarnath:/mnt/c/Users/aditi/Downloads/crowdfunding_cc_2/contracts/investme
ntcc$ cd ..
aditidubey@Kedarnath:/mnt/c/Users/aditi/Downloads/crowdfunding_cc_2/contracts$ cd ..
aditidubey@Kedarnath:/mnt/c/Users/aditi/Downloads/crowdfunding_cc_2$ peer lifecycle
chaincode package investmentcc.tar.gz \
  --path ./contracts/investmentcc \
  --lang golang \
  --label investmentcc_1
aditidubey@Kedarnath:/mnt/c/Users/aditi/Downloads/crowdfunding_cc_2$ peer lifecycle
chaincode install investmentcc.tar.gz
2026-02-07 14:58:56.810 UTC 0001 INFO [cli.lifecycle.chaincode] submitInstallProposal ->
Installed remotely: response:<status:200
payload:"\nOinvestmentcc_1:0b366fdfd184bc98f2cee449c7f7631ca4880d5d598b6c532d72
a142c143aa44\022\016investmentcc_1" >
2026-02-07 14:58:56.813 UTC 0002 INFO [cli.lifecycle.chaincode] submitInstallProposal ->
Chaincode code package identifier:
investmentcc_1:0b366fdfd184bc98f2cee449c7f7631ca4880d5d598b6c532d72a142c143aa4
4
aditidubey@Kedarnath:/mnt/c/Users/aditi/Downloads/crowdfunding_cc_2$ peer lifecycle
chaincode approveformyorg \
  --channelID gov-validation-channel \
  --name governancecc \
  --version 1.0 \
  --sequence 1 \
  --package-id
governancecc_1:8e7e61cdd46e166483b9282a4021b18a6774eea3d55cf8849bf8656e58bbf
574
2026-02-07 15:04:12.387 UTC 0001 INFO [cli.lifecycle.chaincode] setOrdererClient ->
Retrieved channel (gov-validation-channel) orderer endpoint:
orderer-api.127-0-0-1.nip.io:9090
2026-02-07 15:04:12.709 UTC 0002 INFO [chaincodeCmd] ClientWait -> txid
[aed0e277223c1e87d67ead62f8f767fef75968a7e41794b5b688aa4d50dbbbed] committed
with status (VALID) at validatororgpeer-api.127-0-0-1.nip.io:9090
aditidubey@Kedarnath:/mnt/c/Users/aditi/Downloads/crowdfunding_cc_2$ peer lifecycle
chaincode approveformyorg \
  --channelID investment-execution-channel \
  --name investmentcc \
  --version 1.0 \
  --sequence 1 \
  --package-id
investmentcc_1:0b366fdfd184bc98f2cee449c7f7631ca4880d5d598b6c532d72a142c143aa4
4
2026-02-07 15:04:52.414 UTC 0001 INFO [cli.lifecycle.chaincode] setOrdererClient ->
Retrieved channel (investment-execution-channel) orderer endpoint:
orderer-api.127-0-0-1.nip.io:9090
2026-02-07 15:04:52.887 UTC 0002 INFO [chaincodeCmd] ClientWait -> txid
[f5ed4632f1f5b0b68623f7545f60d998c1e5de20942214a4335801f8926889d0] committed
with status (VALID) at validatororgpeer-api.127-0-0-1.nip.io:9090
```

```
aditidubey@Kedarnath:/mnt/c/Users/aditi/Downloads/crowdfunding_cc_2$ peer lifecycle chaincode commit \
  --channelID gov-validation-channel \
  --name governancecc \
  --version 1.0 \
  --sequence 1
2026-02-07 15:09:00.921 UTC 0001 INFO [cli.lifecycle.chaincode] setOrdererClient -> Retrieved channel (gov-validation-channel) orderer endpoint: orderer-api.127-0-0-1.nip.io:9090
2026-02-07 15:09:01.218 UTC 0002 INFO [chaincodeCmd] ClientWait -> txid [69314cb6a4a12d971ca15b93b64486b9658fc53ded471798167a2847a7965b8b] committed with status (VALID) at validatororgpeer-api.127-0-0-1.nip.io:9090
aditidubey@Kedarnath:/mnt/c/Users/aditi/Downloads/crowdfunding_cc_2$ peer lifecycle chaincode commit \
  --channelID investment-execution-channel \
  --name investmentcc \
  --version 1.0 \
  --sequence 1
2026-02-07 15:09:18.652 UTC 0001 INFO [cli.lifecycle.chaincode] setOrdererClient -> Retrieved channel (investment-execution-channel) orderer endpoint: orderer-api.127-0-0-1.nip.io:9090
2026-02-07 15:09:19.132 UTC 0002 INFO [chaincodeCmd] ClientWait -> txid [2be6d1e494c478e032340c74fb53da62ff839424b4eb7bb687c64bea0b3688f0] committed with status (VALID) at validatororgpeer-api.127-0-0-1.nip.io:9090
aditidubey@Kedarnath:/mnt/c/Users/aditi/Downloads/crowdfunding_cc_2$ peer lifecycle chaincode querycommitted \
  --channelID gov-validation-channel
Committed chaincode definitions on channel 'gov-validation-channel':
Name: governancecc, Version: 1.0, Sequence: 1, Endorsement Plugin: escc, Validation Plugin: vscc
aditidubey@Kedarnath:/mnt/c/Users/aditi/Downloads/crowdfunding_cc_2$ docker ps | grep dev
aditidubey@Kedarnath:/mnt/c/Users/aditi/Downloads/crowdfunding_cc_2$ 🚀 STEP 2 — Simple Query Test (Governance)
aditidubey@Kedarnath:/mnt/c/Users/aditi/Downloads/crowdfunding_cc_2$ docker ps | grep dev

aditidubey@Kedarnath:/mnt/c/Users/aditi/Downloads/crowdfunding_cc_2$ peer chaincode invoke \
  -C gov-validation-channel \
  -n governancecc \
  -c '{"function":"SubmitCampaign","Args":["CAMP001","AI Startup","Pending"]}'
2026-02-07 15:13:28.426 UTC 0001 INFO [chaincodeCmd] InitCmdFactory -> Retrieved channel (gov-validation-channel) orderer endpoint: orderer-api.127-0-0-1.nip.io:9090
2026-02-07 15:13:28.542 UTC 0002 INFO [chaincodeCmd] chaincodeInvokeOrQuery -> Chaincode invoke successful. result: status:200
aditidubey@Kedarnath:/mnt/c/Users/aditi/Downloads/crowdfunding_cc_2$ peer chaincode query \
```

```
  -C gov-validation-channel \
  -n governancecc \
  -c '{"function":"GetAllCampaigns","Args":[]}'
Error: endorsement failure during query. response: status:500 message:"Function
GetAllCampaigns not found in contract GovernanceContract"
aditidubey@Kedarnath:/mnt/c/Users/aditi/Downloads/crowdfunding_cc_2$ peer chaincode
invoke \
  -C gov-validation-channel \
  -C gov-validation-channel \
  -n governancecc \
  -c '{"function":"GetAllCampaigns","Args":[]}'
Error: endorsement failure during query. response: status:500 message:"Function
GetAllCampaigns not found in contract GovernanceContract"
aditidubey@Kedarnath:/mnt/c/Users/aditi/Downloads/crowdfunding_cc_2$ peer chaincode
invoke \
  -C gov-validation-channel \
"
aditidubey@Kedarnath:/mnt/c/Users/aditi/Downloads/crowdfunding_cc_2$ peer chaincode
invoke \
  -C gov-validation-channel \
aditidubey@Kedarnath:/mnt/c/Users/aditi/Downloads/crowdfunding_cc_2$ peer chaincode
invoke \
  -C gov-validation-channel \
  -C gov-validation-channel \
  -n governancecc \
  -c '{"function":"ApproveCampaign","Args":["CAMP001","Approved"]}'
2026-02-07 15:16:59.460 UTC 0001 INFO [chaincodeCmd] InitCmdFactory -> Retrieved
channel (gov-validation-channel) orderer endpoint: orderer-api.127-0-0-1.nip.io:9090
2026-02-07 15:16:59.489 UTC 0002 INFO [chaincodeCmd] chaincodeInvokeOrQuery ->
Chaincode invoke successful. result: status:200
aditidubey@Kedarnath:/mnt/c/Users/aditi/Downloads/crowdfunding_cc_2$ peer chaincode
query \
  -C gov-validation-channel \
  -n governancecc \
  -c '{"function":"GetCampaign","Args":["CAMP001"]}'
Error: endorsement failure during query. response: status:500 message:"Function
GetCampaign not found in contract GovernanceContract"
aditidubey@Kedarnath:/mnt/c/Users/aditi/Downloads/crowdfunding_cc_2$
```

Platform terminal:

```
aditidubey@Kedarnath:/mnt/c/Users/aditi/Downloads/crowdfunding_cc_2$
export CORE_PEER_LOCALMSPID=PlatformOrgMSP
export
CORE_PEER_MSPCONFIGPATH=/mnt/c/Users/aditi/Downloads/crowdfunding_cc_2/_msp/
PlatformOrg/platformorgadmin/msp
export CORE_PEER_ADDRESS=platformorgpeer-api.127-0-0-1.nip.io:9090
```

```
aditidubey@Kedarnath:/mnt/c/Users/aditi/Downloads/crowdfunding_cc_2$  export
PATH=$PATH:${PWD}/bin
    export FABRIC_CFG_PATH=${PWD}/config
aditidubey@Kedarnath:/mnt/c/Users/aditi/Downloads/crowdfunding_cc_2$ peer lifecycle
chaincode install governancecc.tar.gz
peer lifecycle chaincode install investmentcc.tar.gz
2026-02-07 15:00:19.087 UTC 0001 INFO [cli.lifecycle.chaincode] submitInstallProposal ->
Installed remotely: response:<status:200
payload:"\nOgovernancecc_1:8e7e61cdd46e166483b9282a4021b18a6774eea3d55cf8849b
f8656e58bbf574\022\016governancecc_1" >
2026-02-07 15:00:19.088 UTC 0002 INFO [cli.lifecycle.chaincode] submitInstallProposal ->
Chaincode code package identifier:
governancecc_1:8e7e61cdd46e166483b9282a4021b18a6774eea3d55cf8849bf8656e58bbf
574
2026-02-07 15:00:24.901 UTC 0001 INFO [cli.lifecycle.chaincode] submitInstallProposal ->
Installed remotely: response:<status:200
payload:"\nOinvestmentcc_1:0b366fdfd184bc98f2cee449c7f7631ca4880d5d598b6c532d72
a142c143aa44\022\016investmentcc_1" >
2026-02-07 15:00:24.902 UTC 0002 INFO [cli.lifecycle.chaincode] submitInstallProposal ->
Chaincode code package identifier:
investmentcc_1:0b366fdfd184bc98f2cee449c7f7631ca4880d5d598b6c532d72a142c143aa4
4
aditidubey@Kedarnath:/mnt/c/Users/aditi/Downloads/crowdfunding_cc_2$ peer lifecycle
chaincode approveformyorg \
  --channelID gov-validation-channel \
  --name governancecc \
  --version 1.0 \
  --sequence 1 \
  --package-id
governancecc_1:8e7e61cdd46e166483b9282a4021b18a6774eea3d55cf8849bf8656e58bbf
574
2026-02-07 15:04:25.335 UTC 0001 INFO [cli.lifecycle.chaincode] setOrdererClient ->
Retrieved channel (gov-validation-channel) orderer endpoint:
orderer-api.127-0-0-1.nip.io:9090
2026-02-07 15:04:25.677 UTC 0002 INFO [chaincodeCmd] ClientWait -> txid
[d4669f2d7dc79becce06ae77da2ada33a225306f05839fd8f302789ae15b3a75] committed
with status (VALID) at platformorgpeer-api.127-0-0-1.nip.io:9090
aditidubey@Kedarnath:/mnt/c/Users/aditi/Downloads/crowdfunding_cc_2$ peer lifecycle
chaincode checkcommitreadiness \
  --channelID gov-validation-channel \
  --name governancecc \
  --version 1.0 \
  --sequence 1 \
  --output json
{
        "approvals": {
                "PlatformOrgMSP": true,
                "StartupOrgMSP": true,
```

```
            "ValidatorOrgMSP": true
        }
}
{
        "approvals": {
            "PlatformOrgMSP": true,
            "StartupOrgMSP": true,
            "ValidatorOrgMSP": true
        }
}
            "ValidatorOrgMSP": true
        }
}
        }
}
}
aditidubey@Kedarnath:/mnt/c/Users/aditi/Downloads/crowdfunding_cc_2$ peer lifecycle
chaincode approveformyorg \
  --channelID investment-execution-channel \
  --name investmentcc \
  --version 1.0 \
  --sequence 1 \
  --package-id
investmentcc_1:0b366fdfd184bc98f2cee449c7f7631ca4880d5d598b6c532d72a142c143aa4
4
2026-02-07 15:05:09.139 UTC 0001 INFO [cli.lifecycle.chaincode] setOrdererClient ->
Retrieved channel (investment-execution-channel) orderer endpoint:
orderer-api.127-0-0-1.nip.io:9090
2026-02-07 15:05:09.537 UTC 0002 INFO [chaincodeCmd] ClientWait -> txid
[c6811351bcc5adb7a51951d4eec014b10dba6d534be1fefc2b4faba9ee47f948] committed
with status (VALID) at platformorgpeer-api.127-0-0-1.nip.io:9090
aditidubey@Kedarnath:/mnt/c/Users/aditi/Downloads/crowdfunding_cc_2$
```

Investor terminal:

```
aditidubey@Kedarnath:/mnt/c/Users/aditi/Downloads/crowdfunding_cc_2$
export CORE_PEER_LOCALMSPID=InvestorOrgMSP
export
CORE_PEER_MSPCONFIGPATH=/mnt/c/Users/aditi/Downloads/crowdfunding_cc_2/_msp/I
nvestorOrg/investororgadmin/msp
export CORE_PEER_ADDRESS=investororgpeer-api.127-0-0-1.nip.io:9090
aditidubey@Kedarnath:/mnt/c/Users/aditi/Downloads/crowdfunding_cc_2$  export
PATH=$PATH:${PWD}/bin
   export FABRIC_CFG_PATH=${PWD}/config
aditidubey@Kedarnath:/mnt/c/Users/aditi/Downloads/crowdfunding_cc_2$ peer lifecycle
chaincode install investmentcc.tar.gz
2026-02-07 15:00:34.592 UTC 0001 INFO [cli.lifecycle.chaincode] submitInstallProposal ->
Installed remotely: response:<status:200
```

```
payload:"\nOinvestmentcc_1:0b366fdfd184bc98f2cee449c7f7631ca4880d5d598b6c532d72
a142c143aa44\022\016investmentcc_1" >
2026-02-07 15:00:34.594 UTC 0002 INFO [cli.lifecycle.chaincode] submitInstallProposal ->
Chaincode code package identifier:
investmentcc_1:0b366fdfd184bc98f2cee449c7f7631ca4880d5d598b6c532d72a142c143aa4
4
aditidubey@Kedarnath:/mnt/c/Users/aditi/Downloads/crowdfunding_cc_2$ peer lifecycle
chaincode queryinstalled
Installed chaincodes on peer:
Package ID:
investmentcc_1:0b366fdfd184bc98f2cee449c7f7631ca4880d5d598b6c532d72a142c143aa4
4, Label: investmentcc_1
aditidubey@Kedarnath:/mnt/c/Users/aditi/Downloads/crowdfunding_cc_2$ peer lifecycle
chaincode approveformyorg \
  --channelID investment-execution-channel \
  --name investmentcc \
  --version 1.0 \
  --sequence 1 \
  --package-id
investmentcc_1:0b366fdfd184bc98f2cee449c7f7631ca4880d5d598b6c532d72a142c143aa4
4
2026-02-07 15:05:13.502 UTC 0001 INFO [cli.lifecycle.chaincode] setOrdererClient ->
Retrieved channel (investment-execution-channel) orderer endpoint:
orderer-api.127-0-0-1.nip.io:9090
2026-02-07 15:05:13.801 UTC 0002 INFO [chaincodeCmd] ClientWait -> txid
[9011dfa327d9ab12f5a802991977ac136e7247dcb2d2728de08b8d88e25b6f2e] committed
with status (VALID) at investororgpeer-api.127-0-0-1.nip.io:9090
aditidubey@Kedarnath:/mnt/c/Users/aditi/Downloads/crowdfunding_cc_2$ peer lifecycle
chaincode checkcommitreadiness \
  --channelID investment-execution-channel \
  --name investmentcc \
  --version 1.0 \
  --sequence 1 \
  --output json
{
        "approvals": {
                "InvestorOrgMSP": true,
                "PlatformOrgMSP": true,
                "StartupOrgMSP": true,
                "ValidatorOrgMSP": true
        }
}
aditidubey@Kedarnath:/mnt/c/Users/aditi/Downloads/crowdfunding_cc_2$ peer chaincode
invoke \
  -C investment-execution-channel \
  -n investmentcc \
  -c '{"function":"CreateInvestor","Args":["INV001","Alice"]}'
```

```
2026-02-07 15:17:29.609 UTC 0001 INFO [chaincodeCmd] InitCmdFactory -> Retrieved
channel (investment-execution-channel) orderer endpoint: orderer-api.127-0-0-1.nip.io:9090
Error: endorsement failure during invoke. response: status:500 message:"Function
CreateInvestor not found in contract InvestmentContract"
aditidubey@Kedarnath:/mnt/c/Users/aditi/Downloads/crowdfunding_cc_2$ peer chaincode
invoke \
  -C investment-execution-channel \
  -n investmentcc \
  -c '{"function":"InvestFunds","Args":["INV001","CAMP001","5000"]}'
2026-02-07 15:18:13.725 UTC 0001 INFO [chaincodeCmd] InitCmdFactory -> Retrieved
channel (investment-execution-channel) orderer endpoint: orderer-api.127-0-0-1.nip.io:9090
Error: endorsement failure during invoke. response: status:500 message:"Incorrect number
of params. Expected 4, received 3"
aditidubey@Kedarnath:/mnt/c/Users/aditi/Downloads/crowdfunding_cc_2$ peer chaincode
invoke   -C investment-execution-channel   -n investmentcc   -c
'{"function":"InvestFunds","Args":["INV001","CAMP001","5000"]}'
  -n investmentcc \
  -c '{"function":"InvestFunds","Args":["INV001","CAMP001","5000"]}'
2026-02-07 15:18:13.725 UTC 0001 INFO [chaincodeCmd] InitCmdFactory -> Retrieved
channel (investment-execution-channel) orderer endpoint: orderer-api.127-0-0-1.nip.io:9090
Error: endorsement failure during invoke. response: status:500 message:"Incorrect number
of params. Expected 4, received 3"
aditidubey@Kedarnath:/mnt/c/Users/aditi/Downloads/crowdfunding_cc_2$ peer chaincode
invoke   -C investment-execution-channel   -n investmentcc   -c
'{"function":"InvestFunds","Args":["INV001","CAMP001","5000"]}'
Error: endorsement failure during invoke. response: status:500 message:"Incorrect number
of params. Expected 4, received 3"
aditidubey@Kedarnath:/mnt/c/Users/aditi/Downloads/crowdfunding_cc_2$ peer chaincode
invoke   -C investment-execution-channel   -n investmentcc   -c
'{"function":"InvestFunds","Args":["INV001","CAMP001","5000"]}'
aditidubey@Kedarnath:/mnt/c/Users/aditi/Downloads/crowdfunding_cc_2$ peer chaincode
invoke   -C investment-execution-channel   -n investmentcc   -c
'{"function":"InvestFunds","Args":["INV001","CAMP001","5000"]}'
mentcc   -c '{"function":"InvestFunds","Args":["INV001","CAMP001","5000"]}'
2026-02-07 15:20:07.055 UTC 0001 INFO [chaincodeCmd] InitCmdFactory -> Retrieved
channel (investment-execution-channel) orderer endpoint: orderer-api.127-0-0-1.nip.io:9090
Error: endorsement failure during invoke. response: status:500 message:"Incorrect number
of params. Expected 4, received 3"
aditidubey@Kedarnath:/mnt/c/Users/aditi/Downloads/crowdfunding_cc_2$ peer chaincode
invoke \
  -C investment-execution-channel \
  -n investmentcc \
  -c '{"function":"InvestFunds","Args":["INVEST001","CAMP001","INV001","5000"]}'
2026-02-07 15:20:19.044 UTC 0001 INFO [chaincodeCmd] InitCmdFactory -> Retrieved
channel (investment-execution-channel) orderer endpoint: orderer-api.127-0-0-1.nip.io:9090
2026-02-07 15:20:19.219 UTC 0002 INFO [chaincodeCmd] chaincodeInvokeOrQuery ->
Chaincode invoke successful. result: status:200
aditidubey@Kedarnath:/mnt/c/Users/aditi/Downloads/crowdfunding_cc_2$
```

Now backend testing route file app.js :
Prerequisite installations commands :

```
aditidubey@Kedarnath:/mnt/c/Users/aditi/Downloads/crowdfunding_cc_2$ mkdir backend
aditidubey@Kedarnath:/mnt/c/Users/aditi/Downloads/crowdfunding_cc_2$ cd backend
aditidubey@Kedarnath:/mnt/c/Users/aditi/Downloads/crowdfunding_cc_2/backend$ npm init
-y
Wrote to /mnt/c/Users/aditi/Downloads/crowdfunding_cc_2/backend/package.json:

{
  "name": "backend",
  "version": "1.0.0",
  "main": "index.js",
  "scripts": {
    "test": "echo \"Error: no test specified\" && exit 1"
  },
  "keywords": [],
  "author": "",
  "license": "ISC",
  "description": ""
}



aditidubey@Kedarnath:/mnt/c/Users/aditi/Downloads/crowdfunding_cc_2/backend$ npm
install express body-parser cors

added 67 packages, and audited 68 packages in 8s

23 packages are looking for funding
  run `npm fund` for details

found 0 vulnerabilities
aditidubey@Kedarnath:/mnt/c/Users/aditi/Downloads/crowdfunding_cc_2/backend$ npm
install @hyperledger/fabric-gateway

added 39 packages, and audited 107 packages in 33s

28 packages are looking for funding
  run `npm fund` for details

found 0 vulnerabilities
aditidubey@Kedarnath:/mnt/c/Users/aditi/Downloads/crowdfunding_cc_2/backend$ npm
install grpc
npm warn deprecated inflight@1.0.6: This module is not supported, and leaks memory. Do
not use it. Check out lru-cache if you want a good and tested way to coalesce async
requests by a key value, which is much more comprehensive and powerful.
npm warn deprecated are-we-there-yet@2.0.0: This package is no longer supported.
```

```
npm warn deprecated rimraf@3.0.2: Rimraf versions prior to v4 are no longer supported
npm warn deprecated npmlog@5.0.1: This package is no longer supported.
npm warn deprecated glob@7.2.3: Old versions of glob are not supported, and contain
widely publicized security vulnerabilities, which have been fixed in the current version.
Please update. Support for old versions may be purchased (at exorbitant rates) by
contacting i@izs.me
npm warn deprecated lodash.clone@4.5.0: This package is deprecated. Use
structuredClone instead.
npm warn deprecated gauge@3.0.2: This package is no longer supported.
npm warn deprecated tar@6.2.1: Old versions of tar are not supported, and contain widely
publicized security vulnerabilities, which have been fixed in the current version. Please
update. Support for old versions may be purchased (at exorbitant rates) by contacting
i@izs.me
npm warn deprecated grpc@1.24.11: This library will not receive further updates other than
security fixes. We recommend using @grpc/grpc-js instead.
npm warn cleanup Failed to remove some directories [
npm warn cleanup   [
npm warn cleanup
'/mnt/c/Users/aditi/Downloads/crowdfunding_cc_2/backend/node_modules/tr46',
npm warn cleanup     [Error: EACCES: permission denied, rmdir
'/mnt/c/Users/aditi/Downloads/crowdfunding_cc_2/backend/node_modules/tr46/lib'] {
npm warn cleanup       errno: -13,
npm warn cleanup       code: 'EACCES',
npm warn cleanup       syscall: 'rmdir',
npm warn cleanup       path:
'/mnt/c/Users/aditi/Downloads/crowdfunding_cc_2/backend/node_modules/tr46/lib'
npm warn cleanup     }
npm warn cleanup   ]
npm warn cleanup ]
npm error code 1
npm error path
/mnt/c/Users/aditi/Downloads/crowdfunding_cc_2/backend/node_modules/grpc
npm error command failed
npm error command sh -c node-pre-gyp install --fallback-to-build --library=static_library
npm error make: Entering directory
'/mnt/c/Users/aditi/Downloads/crowdfunding_cc_2/backend/node_modules/grpc/build'
npm error   CXX(target) Release/obj.target/grpc/deps/grpc/src/core/lib/surface/init.o
npm error make: Leaving directory
'/mnt/c/Users/aditi/Downloads/crowdfunding_cc_2/backend/node_modules/grpc/build'
npm error Failed to execute '/usr/bin/node
/usr/lib/node_modules/npm/node_modules/node-gyp/bin/node-gyp.js build --fallback-to-build
--library=static_library
--module=/mnt/c/Users/aditi/Downloads/crowdfunding_cc_2/backend/node_modules/grpc/sr
c/node/extension_binary/node-v115-linux-x64-glibc/grpc_node.node
--module_name=grpc_node
--module_path=/mnt/c/Users/aditi/Downloads/crowdfunding_cc_2/backend/node_modules/gr
pc/src/node/extension_binary/node-v115-linux-x64-glibc --napi_version=9
--node_abi_napi=napi --napi_build_version=0 --node_napi_label=node-v115' (1)
```

```
npm error node-pre-gyp info it worked if it ends with ok
npm error node-pre-gyp info using node-pre-gyp@1.0.11
npm error node-pre-gyp info using node@20.20.0 | linux | x64
npm error node-pre-gyp info check checked for
"/mnt/c/Users/aditi/Downloads/crowdfunding_cc_2/backend/node_modules/grpc/src/node/ext
ension_binary/node-v115-linux-x64-glibc/grpc_node.node" (not found)
npm error node-pre-gyp http GET
https://node-precompiled-binaries.grpc.io/grpc/v1.24.11/node-v115-linux-x64-glibc.tar.gz
npm error node-pre-gyp ERR! install response status 404 Not Found on
https://node-precompiled-binaries.grpc.io/grpc/v1.24.11/node-v115-linux-x64-glibc.tar.gz
npm error node-pre-gyp WARN Pre-built binaries not installable for grpc@1.24.11 and
node@20.20.0 (node-v115 ABI, glibc) (falling back to source compile with node-gyp)
npm error node-pre-gyp WARN Hit error response status 404 Not Found on
https://node-precompiled-binaries.grpc.io/grpc/v1.24.11/node-v115-linux-x64-glibc.tar.gz
npm error gyp info it worked if it ends with ok
npm error gyp info using node-gyp@10.1.0
npm error gyp info using node@20.20.0 | linux | x64
npm error gyp info ok
npm error gyp info it worked if it ends with ok
npm error gyp info using node-gyp@10.1.0
npm error gyp info using node@20.20.0 | linux | x64
npm error gyp info find Python using Python version 3.12.3 found at "/usr/bin/python3"
npm error gyp info spawn /usr/bin/python3
npm error gyp info spawn args [
npm error gyp info spawn args
'/usr/lib/node_modules/npm/node_modules/node-gyp/gyp/gyp_main.py',
npm error gyp info spawn args 'binding.gyp',
npm error gyp info spawn args '-f',
npm error gyp info spawn args 'make',
npm error gyp info spawn args '-I',
npm error gyp info spawn args
'/mnt/c/Users/aditi/Downloads/crowdfunding_cc_2/backend/node_modules/grpc/build/config.
gypi',
npm error gyp info spawn args '-I',
npm error gyp info spawn args
'/usr/lib/node_modules/npm/node_modules/node-gyp/addon.gypi',
npm error gyp info spawn args '-I',
npm error gyp info spawn args
'/home/aditidubey/.cache/node-gyp/20.20.0/include/node/common.gypi',
npm error gyp info spawn args '-Dlibrary=shared_library',
npm error gyp info spawn args '-Dvisibility=default',
npm error gyp info spawn args
'-Dnode_root_dir=/home/aditidubey/.cache/node-gyp/20.20.0',
npm error gyp info spawn args
'-Dnode_gyp_dir=/usr/lib/node_modules/npm/node_modules/node-gyp',
npm error gyp info spawn args
'-Dnode_lib_file=/home/aditidubey/.cache/node-gyp/20.20.0/<(target_arch)/node.lib',
```

```
npm error gyp info spawn args
'-Dmodule_root_dir=/mnt/c/Users/aditi/Downloads/crowdfunding_cc_2/backend/node_modul
es/grpc',
npm error gyp info spawn args '-Dnode_engine=v8',
npm error gyp info spawn args '--depth=.',
npm error gyp info spawn args '--no-parallel',
npm error gyp info spawn args '--generator-output',
npm error gyp info spawn args 'build',
npm error gyp info spawn args '-Goutput_dir=.'
npm error gyp info spawn args ]
npm error gyp info ok
npm error gyp info it worked if it ends with ok
npm error gyp info using node-gyp@10.1.0
npm error gyp info using node@20.20.0 | linux | x64
npm error gyp info spawn make
npm error gyp info spawn args [ 'BUILDTYPE=Release', '-C', 'build' ]
npm error make: g++: No such file or directory
npm error make: *** [grpc.target.mk:540:
Release/obj.target/grpc/deps/grpc/src/core/lib/surface/init.o] Error 127
npm error gyp ERR! build error
npm error gyp ERR! stack Error: `make` failed with exit code: 2
npm error gyp ERR! stack at ChildProcess.<anonymous>
(/usr/lib/node_modules/npm/node_modules/node-gyp/lib/build.js:209:23)
npm error gyp ERR! System Linux 6.6.87.2-microsoft-standard-WSL2
npm error gyp ERR! command "/usr/bin/node"
"/usr/lib/node_modules/npm/node_modules/node-gyp/bin/node-gyp.js" "build"
"--fallback-to-build" "--library=static_library"
"--module=/mnt/c/Users/aditi/Downloads/crowdfunding_cc_2/backend/node_modules/grpc/sr
c/node/extension_binary/node-v115-linux-x64-glibc/grpc_node.node"
"--module_name=grpc_node"
"--module_path=/mnt/c/Users/aditi/Downloads/crowdfunding_cc_2/backend/node_modules/g
rpc/src/node/extension_binary/node-v115-linux-x64-glibc" "--napi_version=9"
"--node_abi_napi=napi" "--napi_build_version=0" "--node_napi_label=node-v115"
npm error gyp ERR! cwd
/mnt/c/Users/aditi/Downloads/crowdfunding_cc_2/backend/node_modules/grpc
npm error gyp ERR! node -v v20.20.0
npm error gyp ERR! node-gyp -v v10.1.0
npm error gyp ERR! not ok
npm error node-pre-gyp ERR! build error
npm error node-pre-gyp ERR! stack Error: Failed to execute '/usr/bin/node
/usr/lib/node_modules/npm/node_modules/node-gyp/bin/node-gyp.js build --fallback-to-build
--library=static_library
--module=/mnt/c/Users/aditi/Downloads/crowdfunding_cc_2/backend/node_modules/grpc/sr
c/node/extension_binary/node-v115-linux-x64-glibc/grpc_node.node
--module_name=grpc_node
--module_path=/mnt/c/Users/aditi/Downloads/crowdfunding_cc_2/backend/node_modules/gr
pc/src/node/extension_binary/node-v115-linux-x64-glibc --napi_version=9
--node_abi_napi=napi --napi_build_version=0 --node_napi_label=node-v115' (1)
```

```
npm error node-pre-gyp ERR! stack     at ChildProcess.<anonymous>
(/mnt/c/Users/aditi/Downloads/crowdfunding_cc_2/backend/node_modules/@mapbox/node-
pre-gyp/lib/util/compile.js:89:23)
npm error node-pre-gyp ERR! stack     at ChildProcess.emit (node:events:524:28)
npm error node-pre-gyp ERR! stack     at maybeClose (node:internal/child_process:1104:16)
npm error node-pre-gyp ERR! stack     at ChildProcess._handle.onexit
(node:internal/child_process:304:5)
npm error node-pre-gyp ERR! System Linux 6.6.87.2-microsoft-standard-WSL2
npm error node-pre-gyp ERR! command "/usr/bin/node"
"/mnt/c/Users/aditi/Downloads/crowdfunding_cc_2/backend/node_modules/.bin/node-pre-gy
p" "install" "--fallback-to-build" "--library=static_library"
npm error node-pre-gyp ERR! cwd
/mnt/c/Users/aditi/Downloads/crowdfunding_cc_2/backend/node_modules/grpc
npm error node-pre-gyp ERR! node -v v20.20.0
npm error node-pre-gyp ERR! node-pre-gyp -v v1.0.11
npm error node-pre-gyp ERR! not ok
npm error A complete log of this run can be found in:
/home/aditidubey/.npm/_logs/2026-02-07T15_38_14_524Z-debug-0.log
aditidubey@Kedarnath:/mnt/c/Users/aditi/Downloads/crowdfunding_cc_2/backend$ npm
install grpc
npm warn deprecated inflight@1.0.6: This module is not supported, and leaks memory. Do
not use it. Check out lru-cache if you want a good and tested way to coalesce async
requests by a key value, which is much more comprehensive and powerful.
npm warn deprecated npmlog@5.0.1: This package is no longer supported.
npm warn deprecated lodash.clone@4.5.0: This package is deprecated. Use
structuredClone instead.
npm warn deprecated rimraf@3.0.2: Rimraf versions prior to v4 are no longer supported
npm warn deprecated are-we-there-yet@2.0.0: This package is no longer supported.
npm warn deprecated glob@7.2.3: Old versions of glob are not supported, and contain
widely publicized security vulnerabilities, which have been fixed in the current version.
Please update. Support for old versions may be purchased (at exorbitant rates) by
contacting i@izs.me
npm warn deprecated gauge@3.0.2: This package is no longer supported.
npm warn deprecated tar@6.2.1: Old versions of tar are not supported, and contain widely
publicized security vulnerabilities, which have been fixed in the current version. Please
update. Support for old versions may be purchased (at exorbitant rates) by contacting
i@izs.me
npm warn deprecated grpc@1.24.11: This library will not receive further updates other than
security fixes. We recommend using @grpc/grpc-js instead.
npm error code 1
npm error path
/mnt/c/Users/aditi/Downloads/crowdfunding_cc_2/backend/node_modules/grpc
npm error command failed
npm error command sh -c node-pre-gyp install --fallback-to-build --library=static_library
npm error make: Entering directory
'/mnt/c/Users/aditi/Downloads/crowdfunding_cc_2/backend/node_modules/grpc/build'
npm error   CXX(target) Release/obj.target/grpc/deps/grpc/src/core/lib/surface/init.o
```

```
npm error make: Leaving directory
'/mnt/c/Users/aditi/Downloads/crowdfunding_cc_2/backend/node_modules/grpc/build'
npm error Failed to execute '/usr/bin/node
/usr/lib/node_modules/npm/node_modules/node-gyp/bin/node-gyp.js build --fallback-to-build
--library=static_library
--module=/mnt/c/Users/aditi/Downloads/crowdfunding_cc_2/backend/node_modules/grpc/sr
c/node/extension_binary/node-v115-linux-x64-glibc/grpc_node.node
--module_name=grpc_node
--module_path=/mnt/c/Users/aditi/Downloads/crowdfunding_cc_2/backend/node_modules/gr
pc/src/node/extension_binary/node-v115-linux-x64-glibc --napi_version=9
--node_abi_napi=napi --napi_build_version=0 --node_napi_label=node-v115' (1)
npm error node-pre-gyp info it worked if it ends with ok
npm error node-pre-gyp info using node-pre-gyp@1.0.11
npm error node-pre-gyp info using node@20.20.0 | linux | x64
npm error node-pre-gyp info check checked for
"/mnt/c/Users/aditi/Downloads/crowdfunding_cc_2/backend/node_modules/grpc/src/node/ext
ension_binary/node-v115-linux-x64-glibc/grpc_node.node" (not found)
npm error node-pre-gyp http GET
https://node-precompiled-binaries.grpc.io/grpc/v1.24.11/node-v115-linux-x64-glibc.tar.gz
npm error node-pre-gyp ERR! install response status 404 Not Found on
https://node-precompiled-binaries.grpc.io/grpc/v1.24.11/node-v115-linux-x64-glibc.tar.gz
npm error node-pre-gyp WARN Pre-built binaries not installable for grpc@1.24.11 and
node@20.20.0 (node-v115 ABI, glibc) (falling back to source compile with node-gyp)
npm error node-pre-gyp WARN Hit error response status 404 Not Found on
https://node-precompiled-binaries.grpc.io/grpc/v1.24.11/node-v115-linux-x64-glibc.tar.gz
npm error gyp info it worked if it ends with ok
npm error gyp info using node-gyp@10.1.0
npm error gyp info using node@20.20.0 | linux | x64
npm error gyp info ok
npm error gyp info it worked if it ends with ok
npm error gyp info using node-gyp@10.1.0
npm error gyp info using node@20.20.0 | linux | x64
npm error gyp info find Python using Python version 3.12.3 found at "/usr/bin/python3"
npm error gyp info spawn /usr/bin/python3
npm error gyp info spawn args [
npm error gyp info spawn args
'/usr/lib/node_modules/npm/node_modules/node-gyp/gyp/gyp_main.py',
npm error gyp info spawn args 'binding.gyp',
npm error gyp info spawn args '-f',
npm error gyp info spawn args 'make',
npm error gyp info spawn args '-I',
npm error gyp info spawn args
'/mnt/c/Users/aditi/Downloads/crowdfunding_cc_2/backend/node_modules/grpc/build/config.
gypi',
npm error gyp info spawn args '-I',
npm error gyp info spawn args
'/usr/lib/node_modules/npm/node_modules/node-gyp/addon.gypi',
npm error gyp info spawn args '-I',
```

```
npm error gyp info spawn args
'/home/aditidubey/.cache/node-gyp/20.20.0/include/node/common.gypi',
npm error gyp info spawn args '-Dlibrary=shared_library',
npm error gyp info spawn args '-Dvisibility=default',
npm error gyp info spawn args
'-Dnode_root_dir=/home/aditidubey/.cache/node-gyp/20.20.0',
npm error gyp info spawn args
'-Dnode_gyp_dir=/usr/lib/node_modules/npm/node_modules/node-gyp',
npm error gyp info spawn args
'-Dnode_lib_file=/home/aditidubey/.cache/node-gyp/20.20.0/<(target_arch)/node.lib',
npm error gyp info spawn args
'-Dmodule_root_dir=/mnt/c/Users/aditi/Downloads/crowdfunding_cc_2/backend/node_modul
es/grpc',
npm error gyp info spawn args '-Dnode_engine=v8',
npm error gyp info spawn args '--depth=.',
npm error gyp info spawn args '--no-parallel',
npm error gyp info spawn args '--generator-output',
npm error gyp info spawn args 'build',
npm error gyp info spawn args '-Goutput_dir=.'
npm error gyp info spawn args ]
npm error gyp info ok
npm error gyp info it worked if it ends with ok
npm error gyp info using node-gyp@10.1.0
npm error gyp info using node@20.20.0 | linux | x64
npm error gyp info spawn make
npm error gyp info spawn args [ 'BUILDTYPE=Release', '-C', 'build' ]
npm error make: g++: No such file or directory
npm error make: *** [grpc.target.mk:540:
Release/obj.target/grpc/deps/grpc/src/core/lib/surface/init.o] Error 127
npm error gyp ERR! build error
npm error gyp ERR! stack Error: `make` failed with exit code: 2
npm error gyp ERR! stack at ChildProcess.<anonymous>
(/usr/lib/node_modules/npm/node_modules/node-gyp/lib/build.js:209:23)
npm error gyp ERR! System Linux 6.6.87.2-microsoft-standard-WSL2
npm error gyp ERR! command "/usr/bin/node"
"/usr/lib/node_modules/npm/node_modules/node-gyp/bin/node-gyp.js" "build"
"--fallback-to-build" "--library=static_library"
"--module=/mnt/c/Users/aditi/Downloads/crowdfunding_cc_2/backend/node_modules/grpc/sr
c/node/extension_binary/node-v115-linux-x64-glibc/grpc_node.node"
"--module_name=grpc_node"
"--module_path=/mnt/c/Users/aditi/Downloads/crowdfunding_cc_2/backend/node_modules/g
rpc/src/node/extension_binary/node-v115-linux-x64-glibc" "--napi_version=9"
"--node_abi_napi=napi" "--napi_build_version=0" "--node_napi_label=node-v115"
npm error gyp ERR! cwd
/mnt/c/Users/aditi/Downloads/crowdfunding_cc_2/backend/node_modules/grpc
npm error gyp ERR! node -v v20.20.0
npm error gyp ERR! node-gyp -v v10.1.0
npm error gyp ERR! not ok
```

```
npm error node-pre-gyp ERR! build error
npm error node-pre-gyp ERR! stack Error: Failed to execute '/usr/bin/node
/usr/lib/node_modules/npm/node_modules/node-gyp/bin/node-gyp.js build --fallback-to-build
--library=static_library
--module=/mnt/c/Users/aditi/Downloads/crowdfunding_cc_2/backend/node_modules/grpc/sr
c/node/extension_binary/node-v115-linux-x64-glibc/grpc_node.node
--module_name=grpc_node
--module_path=/mnt/c/Users/aditi/Downloads/crowdfunding_cc_2/backend/node_modules/gr
pc/src/node/extension_binary/node-v115-linux-x64-glibc --napi_version=9
--node_abi_napi=napi --napi_build_version=0 --node_napi_label=node-v115' (1)
npm error node-pre-gyp ERR! stack     at ChildProcess.<anonymous>
(/mnt/c/Users/aditi/Downloads/crowdfunding_cc_2/backend/node_modules/@mapbox/node-
pre-gyp/lib/util/compile.js:89:23)
npm error node-pre-gyp ERR! stack     at ChildProcess.emit (node:events:524:28)
npm error node-pre-gyp ERR! stack     at maybeClose (node:internal/child_process:1104:16)
npm error node-pre-gyp ERR! stack     at ChildProcess._handle.onexit
(node:internal/child_process:304:5)
npm error node-pre-gyp ERR! System Linux 6.6.87.2-microsoft-standard-WSL2
npm error node-pre-gyp ERR! command "/usr/bin/node"
"/mnt/c/Users/aditi/Downloads/crowdfunding_cc_2/backend/node_modules/.bin/node-pre-gy
p" "install" "--fallback-to-build" "--library=static_library"
npm error node-pre-gyp ERR! cwd
/mnt/c/Users/aditi/Downloads/crowdfunding_cc_2/backend/node_modules/grpc
npm error node-pre-gyp ERR! node -v v20.20.0
npm error node-pre-gyp ERR! node-pre-gyp -v v1.0.11
npm error node-pre-gyp ERR! not ok
npm error A complete log of this run can be found in:
/home/aditidubey/.npm/_logs/2026-02-07T15_41_11_917Z-debug-0.log
aditidubey@Kedarnath:/mnt/c/Users/aditi/Downloads/crowdfunding_cc_2/backend$ npm
uninstall grpc

removed 1 package, and audited 107 packages in 7s

28 packages are looking for funding
  run `npm fund` for details

found 0 vulnerabilities
aditidubey@Kedarnath:/mnt/c/Users/aditi/Downloads/crowdfunding_cc_2/backend$ npm
install express body-parser cors
npm install @hyperledger/fabric-gateway

up to date, audited 107 packages in 5s

28 packages are looking for funding
  run `npm fund` for details

found 0 vulnerabilities
```

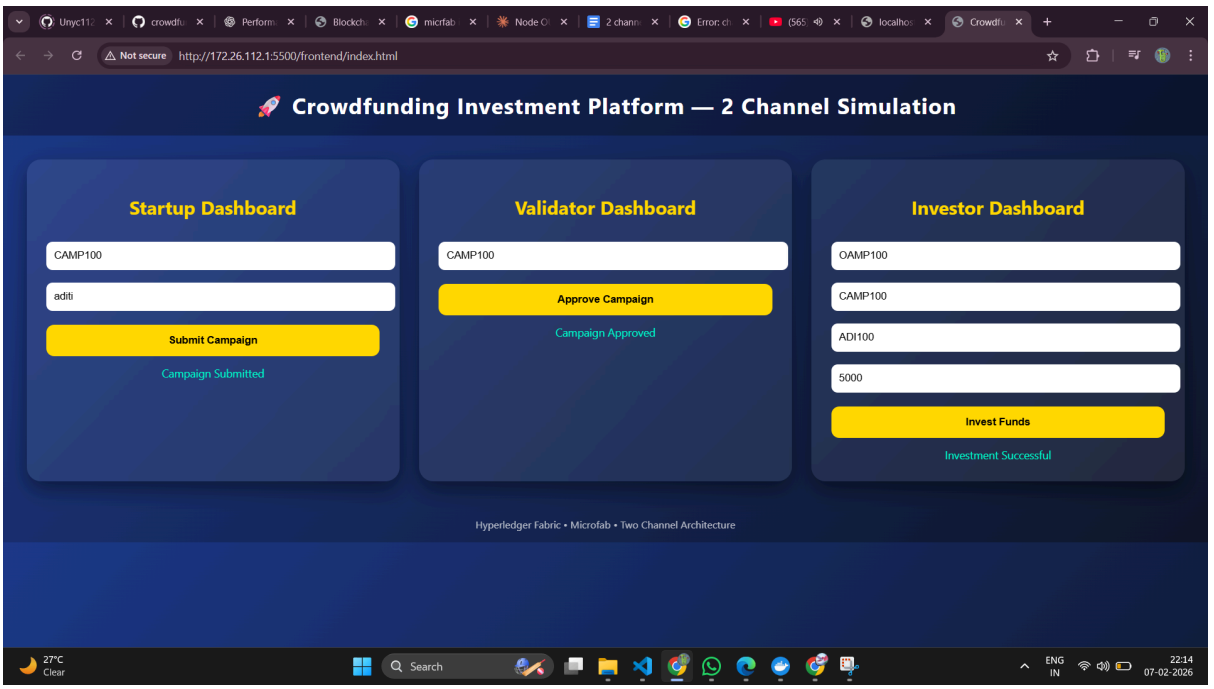
```
up to date, audited 107 packages in 5s

28 packages are looking for funding
  run `npm fund` for details

found 0 vulnerabilities
aditidubey@Kedarnath:/mnt/c/Users/aditi/Downloads/crowdfunding_cc_2/backend$ npm list
--depth=0
backend@1.0.0 /mnt/c/Users/aditi/Downloads/crowdfunding_cc_2/backend
├── @hyperledger/fabric-gateway@1.10.1
├── body-parser@2.2.2
├── cors@2.8.6
└── express@5.2.1

aditidubey@Kedarnath:/mnt/c/Users/aditi/Downloads/crowdfunding_cc_2/backend$ node
app.js
Backend API running on port 3000
^C
aditidubey@Kedarnath:/mnt/c/Users/aditi/Downloads/crowdfunding_cc_2/backend$ curl -X
POST http://localhost:3000/campaign \
-H "Content-Type: application/json" \
-d '{"id":"CAMP500","title":"AI Fund"}'
ENOENT: no such file or directory, open
'/mnt/c/Users/aditi/Downloads/crowdfunding_cc_2/_gateways/connection-profile-startuporg.j
son'aditidubey@Kedarnath:/mnt/c/Users/aditi/Downloads/crowdfunding_cc_2/backend$
nodnode app.js
^C
aditidubey@Kedarnath:/mnt/c/Users/aditi/Downloads/crowdfunding_cc_2/backend$ curl -X
POST http://localhost:3000/campaign \
-H "Content-Type: application/json" \
-d '{"id":"CAMP500","title":"AI Fund"}'
aditidubey@Kedarnath:/mnt/c/Users/aditi/Downloads/crowdfunding_cc_2/backend$
```

To run server: command node app.js

Frontend : index.html (for now it is very minimalistic - no auth)

- Go live

-



Terminal references :

aditidubey@Kedarnath:/mnt/c/Users/aditi/Downloads/crowdfunding_cc_2$ cd backend
aditidubey@Kedarnath:/mnt/c/Users/aditi/Downloads/crowdfunding_cc_2/backend$ curl -X POST http://localhost:3000/campaign \
-H "Content-Type: application/json" \
-d '{"id":"CAMP500","title":"AI Fund"}'
Cannot read properties of undefined (reading 'newFileSystemWallet')aditidubey@Kedarnath:/mnt/c/Users/aditi/Downloads/crowdfunding_cc_2/backend$ npm install fabric-network npm install fabric-network

added 46 packages, and audited 153 packages in 39s

31 packages are looking for funding
  run `npm fund` for details

4 vulnerabilities (2 low, 2 high)

added 46 packages, and audited 153 packages in 39s

31 packages are looking for funding
  run `npm fund` for details

4 vulnerabilities (2 low, 2 high)
  run `npm fund` for details

4 vulnerabilities (2 low, 2 high)

```
4 vulnerabilities (2 low, 2 high)
4 vulnerabilities (2 low, 2 high)

To address all issues, run:
  npm audit fix

Run `npm audit` for details.
aditidubey@Kedarnath:/mnt/c/Users/aditi/Downloads/crowdfunding_cc_2/backend$ node
app.js
Backend API running on port 3000
^C
aditidubey@Kedarnath:/mnt/c/Users/aditi/Downloads/crowdfunding_cc_2/backend$ node
app.js
Backend API running on port 3000
```

complex 2 channel

Port : 9091

Packaging commands :

Gov

```
peer lifecycle chaincode package governancecc_v2.tar.gz \
  --path ./contracts \
  --lang golang \
  --label governancecc_2
```

Inv

```
peer lifecycle chaincode package investmentcc_v2.tar.gz \
  --path ./contracts \
  --lang golang \
  --label investmentcc_2
```

CHAINCODES

FRONTEND

WALLET LOGIC

- Each Startup / Investor has a **wallet asset** on blockchain.
- Wallet = `{ id, ownerId, type, balance }`

Flow:

1. Platform creates wallets → initial balance set.
2. When **Investor invests**:
    - Investor wallet → **balance decreases**
    - Startup wallet → **balance increases**
3. Chaincode checks wallet exists + enough balance.

4. If wallet missing or low balance → transaction fails.
5. Used for:
    - Investments
    - Fee deduction
    - Deletion charges
    - Fund release tracking

GOVERNANCECC.GO

```go
package main

import (
    "crypto/sha256"
    "encoding/hex"
    "encoding/json"
    "fmt"
    "time"

    "github.com/hyperledger/fabric-contract-api-go/contractapi"
)

type GovernanceContract struct {
    contractapi.Contract
}

/* ============================================================
   ====================== MODELS ==============================
============================================================ */

type Campaign struct {
    ID              string `json:"id"`
    Name            string `json:"name"`
    StartupID       string `json:"startupId"`
    Description     string `json:"description"`
    GoalAmount      int    `json:"goalAmount"`
    FundsRaised     int    `json:"fundsRaised"`
    Status          string `json:"status"`
    RiskScore       int    `json:"riskScore"`
    ComplianceScore int    `json:"complianceScore"`
    ApprovalHash    string `json:"approvalHash"`
    CreatedAt       string `json:"createdAt"`
    Deleted         bool   `json:"deleted"`
}

type ValidationRecord struct {
```

```
    CampaignID string `json:"campaignId"`
    Validator  string `json:"validator"`
    Decision   string `json:"decision"`
    Notes      string `json:"notes"`
    Timestamp  string `json:"timestamp"`
}

type Milestone struct {
    ID           string `json:"id"`
    CampaignID   string `json:"campaignId"`
    Title        string `json:"title"`
    ProofHash    string `json:"proofHash"`
    Status       string `json:"status"`
    ApprovedHash string `json:"approvedHash"`
}

type DeletionFeePreview struct {
    EntityId      string `json:"entityId"`
    EntityType    string `json:"entityType"`
    FundsRaised   int    `json:"fundsRaised"`
    FeeAmount     int    `json:"feeAmount"`
    FeePercentage int    `json:"feePercentage"`
    IsFixedFee    bool   `json:"isFixedFee"`
}

/* ============================================================
   ===================== HELPERS ============================
============================================================ */

func hashString(input string) string {
    h := sha256.Sum256([]byte(input))
    return hex.EncodeToString(h[:])
}

func getMSP(ctx contractapi.TransactionContextInterface) (string,
error) {
    return ctx.GetClientIdentity().GetMSPID()
}

/* ============================================================
   ================= CAMPAIGN FLOW ==========================
============================================================ */
```

```go
func (g *GovernanceContract) CreateCampaign(
    ctx contractapi.TransactionContextInterface,
    id, name, startupId, desc string,
    goal int,
) error {

    if b, _ := ctx.GetStub().GetState(id); b != nil {
        return fmt.Errorf("campaign already exists")
    }

    c := Campaign{
        ID:          id,
        Name:        name,
        StartupID:   startupId,
        Description: desc,
        GoalAmount:  goal,
        Status:      "SUBMITTED",
        CreatedAt:   time.Now().Format(time.RFC3339),
    }

    js, _ := json.Marshal(c)
    return ctx.GetStub().PutState(id, js)
}

func (g *GovernanceContract) GetCampaign(
    ctx contractapi.TransactionContextInterface,
    id string,
) (*Campaign, error) {

    b, err := ctx.GetStub().GetState(id)
    if err != nil || b == nil {
        return nil, fmt.Errorf("campaign not found")
    }

    var c Campaign
    json.Unmarshal(b, &c)
    return &c, nil
}

/* ============================================================
   ================= VALIDATION FLOW =======================
============================================================ */
```

```go
func (g *GovernanceContract) SubmitForValidation(
    ctx contractapi.TransactionContextInterface,
    id string,
) error {

    c, err := g.GetCampaign(ctx, id)
    if err != nil {
        return err
    }

    if c.Status != "SUBMITTED" {
        return fmt.Errorf("invalid state")
    }

    c.Status = "UNDER_VALIDATION"
    js, _ := json.Marshal(c)
    return ctx.GetStub().PutState(id, js)
}

func (g *GovernanceContract) RecordValidation(
    ctx contractapi.TransactionContextInterface,
    campaignId, validator, decision, notes string,
) error {

    msp, _ := getMSP(ctx)
    if msp != "ValidatorOrgMSP" {
        return fmt.Errorf("only validator allowed")
    }

    rec := ValidationRecord{
        CampaignID: campaignId,
        Validator:  validator,
        Decision:   decision,
        Notes:      notes,
        Timestamp:  time.Now().Format(time.RFC3339),
    }

    key := "VAL_" + campaignId + "_" + validator
    js, _ := json.Marshal(rec)
    return ctx.GetStub().PutState(key, js)
}

/* ==========================================================
```

```go
   ================= SCORING ================================
=========================================================== */

func (g *GovernanceContract) SetRiskScore(
    ctx contractapi.TransactionContextInterface,
    id string,
    score int,
) error {

    msp, _ := getMSP(ctx)
    if msp != "ValidatorOrgMSP" {
        return fmt.Errorf("validator only")
    }

    c, err := g.GetCampaign(ctx, id)
    if err != nil {
        return err
    }

    c.RiskScore = score
    js, _ := json.Marshal(c)
    return ctx.GetStub().PutState(id, js)
}

func (g *GovernanceContract) SetComplianceScore(
    ctx contractapi.TransactionContextInterface,
    id string,
    score int,
) error {

    msp, _ := getMSP(ctx)
    if msp != "ValidatorOrgMSP" {
        return fmt.Errorf("validator only")
    }

    c, err := g.GetCampaign(ctx, id)
    if err != nil {
        return err
    }

    c.ComplianceScore = score
    js, _ := json.Marshal(c)
    return ctx.GetStub().PutState(id, js)
```

```go
}

/* ============================================================
   ================= APPROVAL ==============================
============================================================ */

func (g *GovernanceContract) ApproveCampaign(
    ctx contractapi.TransactionContextInterface,
    id string,
) (string, error) {

    msp, _ := getMSP(ctx)
    if msp != "ValidatorOrgMSP" {
        return "", fmt.Errorf("validator only")
    }

    c, err := g.GetCampaign(ctx, id)
    if err != nil {
        return "", err
    }

    if c.RiskScore == 0 || c.ComplianceScore == 0 {
        return "", fmt.Errorf("scores missing")
    }

    hashInput := fmt.Sprintf("%s|%d|%d|%s",
        id, c.RiskScore, c.ComplianceScore, time.Now().String())

    c.Status = "APPROVED"
    c.ApprovalHash = hashString(hashInput)

    js, _ := json.Marshal(c)
    ctx.GetStub().PutState(id, js)

    return c.ApprovalHash, nil
}

func (g *GovernanceContract) RejectCampaign(
    ctx contractapi.TransactionContextInterface,
    id string,
) error {

    c, err := g.GetCampaign(ctx, id)
```

```go
    if err != nil {
        return err
    }

    c.Status = "REJECTED"
    js, _ := json.Marshal(c)
    return ctx.GetStub().PutState(id, js)
}

/* ============================================================
   ================= MILESTONES ============================
============================================================ */

func (g *GovernanceContract) SubmitMilestone(
    ctx contractapi.TransactionContextInterface,
    id, campaignId, title, proof string,
) error {

    m := Milestone{
        ID:         id,
        CampaignID: campaignId,
        Title:      title,
        ProofHash:  hashString(proof),
        Status:     "SUBMITTED",
    }

    js,_ := json.Marshal(m)
    return ctx.GetStub().PutState(id, js)
}

func (g *GovernanceContract) ApproveMilestone(
    ctx contractapi.TransactionContextInterface,
    id string,
) (string, error) {

    msp,_ := getMSP(ctx)
    if msp != "ValidatorOrgMSP" {
        return "", fmt.Errorf("validator only")
    }

    b,_ := ctx.GetStub().GetState(id)
    if b == nil {
        return "", fmt.Errorf("milestone missing")
```

```go
    }

    var m Milestone
    json.Unmarshal(b,&m)

    m.Status = "APPROVED"
    m.ApprovedHash = hashString(id + time.Now().String())

    js,_ := json.Marshal(m)
    ctx.GetStub().PutState(id, js)

    return m.ApprovedHash, nil
}

/* ============================================================
   ================= DELETION FEES =========================
============================================================ */

func (g *GovernanceContract) CalculateCampaignDeletionFee(
    ctx contractapi.TransactionContextInterface,
    id string,
) (*DeletionFeePreview,error){

    c,_ := g.GetCampaign(ctx,id)

    p := DeletionFeePreview{
        EntityId: id,
        EntityType: "CAMPAIGN",
        FundsRaised: c.FundsRaised,
    }

    if c.FundsRaised == 0 {
        p.FeeAmount = 100
        p.IsFixedFee = true
    } else {
        p.FeeAmount = (c.FundsRaised * 60)/100
        p.FeePercentage = 60
    }

    return &p,nil
}

func (g *GovernanceContract) DeleteCampaign(
```

```go
    ctx contractapi.TransactionContextInterface,
    id, reason string,
) error {

    c,_ := g.GetCampaign(ctx,id)

    p,_ := g.CalculateCampaignDeletionFee(ctx,id)

    c.Status = "DELETED"
    c.Deleted = true

    js,_ := json.Marshal(c)
    ctx.GetStub().PutState(id, js)

    rec := map[string]interface{}{
        "id": id,
        "reason": reason,
        "fee": p.FeeAmount,
        "time": time.Now().String(),
    }

    rj,_ := json.Marshal(rec)
    return ctx.GetStub().PutState("DEL_"+id,rj)
}

/* ============================================================
   ================= HISTORY ==============================
============================================================ */

func (g *GovernanceContract) GetHistory(
    ctx contractapi.TransactionContextInterface,
    id string,
)([]string,error){

    iter,_ := ctx.GetStub().GetHistoryForKey(id)
    defer iter.Close()

    var out []string
    for iter.HasNext(){
        r,_ := iter.Next()
        out = append(out,string(r.Value))
    }
    return out,nil
```

```go
}

func (g *GovernanceContract) AddFundsRaised(
 ctx contractapi.TransactionContextInterface,
 id string,
 amount int,
) error {

 c, err := g.GetCampaign(ctx, id)
 if err != nil {
  return err
 }

 if c.Deleted {
  return fmt.Errorf("campaign deleted")
 }

 c.FundsRaised += amount

 js,_ := json.Marshal(c)
 return ctx.GetStub().PutState(id, js)
}


/* ============================================================ */

func main(){
    cc,err := contractapi.NewChaincode(&GovernanceContract{})
    if err != nil { panic(err) }
    cc.Start()
}
```

INVESTMENTCC.GO

```go
package main

import (
    "encoding/json"
    "fmt"
    "time"

    "github.com/hyperledger/fabric-contract-api-go/contractapi"
)
```

```go
type InvestmentContract struct {
    contractapi.Contract
}

/* ============================================================
   ======================== MODELS ==========================
============================================================ */

type Wallet struct {
    ID      string `json:"id"`
    OwnerID string `json:"ownerId"`
    Type    string `json:"type"`   // STARTUP / INVESTOR / PLATFORM
    Balance int    `json:"balance"`
}

type Investment struct {
    ID           string `json:"id"`
    CampaignID   string `json:"campaignId"`
    InvestorID   string `json:"investorId"`
    Amount       int    `json:"amount"`
    Status       string `json:"status"` // INVESTED / RELEASED /
REFUNDED / DISPUTED
    ApprovalHash string `json:"approvalHash"`
    MilestoneRef string `json:"milestoneRef"`
    CreatedAt    string `json:"createdAt"`
}

type CampaignFeeTier struct {
    ID          string `json:"id"`
    MinAmount   int    `json:"minAmount"`
    MaxAmount   int    `json:"maxAmount"`
    Percent     int    `json:"percent"`
    Description string `json:"description"`
}

type DisputeFeeTier struct {
    ID          string `json:"id"`
    MinAmount   int    `json:"minAmount"`
    MaxAmount   int    `json:"maxAmount"`
    FlatFee     int    `json:"flatFee"`
    Description string `json:"description"`
}
```

```go
/* ============================================================
   ======================== HELPERS ==========================
============================================================ */

func getMSP(ctx contractapi.TransactionContextInterface) (string,
error) {
    return ctx.GetClientIdentity().GetMSPID()
}

func put(ctx contractapi.TransactionContextInterface, key string, v
interface{}) error {
    js, _ := json.Marshal(v)
    return ctx.GetStub().PutState(key, js)
}

func getWallet(ctx contractapi.TransactionContextInterface, id string)
(*Wallet, error) {
    b, err := ctx.GetStub().GetState(id)
    if err != nil || b == nil {
        return nil, fmt.Errorf("wallet not found")
    }
    var w Wallet
    json.Unmarshal(b, &w)
    return &w, nil
}

func (ic *InvestmentContract) walletKey(ownerID string) string {
    return "WALLET_" + ownerID
}

/* ============================================================
   ======================== WALLET ===========================
============================================================ */

func (ic *InvestmentContract) CreateWallet(
    ctx contractapi.TransactionContextInterface,
    walletID string,
    ownerID string,
    wtype string,
    balance int,
) error {
```

```go
    msp, _ := getMSP(ctx)
    if msp != "PlatformOrgMSP" {
        return fmt.Errorf("only platform can create wallets")
    }

    if b, _ := ctx.GetStub().GetState(walletID); b != nil {
        return fmt.Errorf("wallet exists")
    }

    w := Wallet{
        ID:      walletID,
        OwnerID: ownerID,
        Type:    wtype,
        Balance: balance,
    }

    return put(ctx, walletID, w)
}

func (ic *InvestmentContract) ReadWallet(
    ctx contractapi.TransactionContextInterface,
    walletID string,
) (*Wallet, error) {
    return getWallet(ctx, walletID)
}

func (ic *InvestmentContract) TransferCFT(
    ctx contractapi.TransactionContextInterface,
    fromWallet string,
    toWallet string,
    amount int,
) error {

    from, err := getWallet(ctx, fromWallet)
    if err != nil {
        return err
    }

    to, err := getWallet(ctx, toWallet)
    if err != nil {
        return err
    }
```

```go
    if from.Balance < amount {
        return fmt.Errorf("insufficient CFT")
    }

    from.Balance -= amount
    to.Balance += amount

    put(ctx, fromWallet, from)
    return put(ctx, toWallet, to)
}

/* ============================================================
   ======================== INVEST ==========================
============================================================ */

func (ic *InvestmentContract) InvestmentExists(
    ctx contractapi.TransactionContextInterface,
    id string,
) (bool, error) {
    b, err := ctx.GetStub().GetState(id)
    return b != nil, err
}

func (ic *InvestmentContract) InvestFunds(
    ctx contractapi.TransactionContextInterface,
    invID string,
    campaignID string,
    investorID string,
    amount int,
    approvalHash string,
) error {

    msp, _ := getMSP(ctx)
    if msp != "InvestorOrgMSP" {
        return fmt.Errorf("only investor org can invest")
    }

    if approvalHash == "" {
        return fmt.Errorf("validator approval required")
    }

    exists, _ := ic.InvestmentExists(ctx, invID)
    if exists {
```

```go
        return fmt.Errorf("investment exists")
    }

    investorWallet := ic.walletKey(investorID)
    platformWallet := "WALLET_PLATFORM"

    // debit investor → credit platform escrow
    err := ic.TransferCFT(ctx, investorWallet, platformWallet, amount)
    if err != nil {
        return err
    }

    inv := Investment{
        ID:           invID,
        CampaignID:   campaignID,
        InvestorID:   investorID,
        Amount:       amount,
        Status:       "INVESTED",
        ApprovalHash: approvalHash,
        CreatedAt:    time.Now().Format(time.RFC3339),
    }

    return put(ctx, invID, inv)
}

func (ic *InvestmentContract) ReadInvestment(
    ctx contractapi.TransactionContextInterface,
    id string,
) (*Investment, error) {

    b, err := ctx.GetStub().GetState(id)
    if err != nil || b == nil {
        return nil, fmt.Errorf("investment not found")
    }

    var inv Investment
    json.Unmarshal(b, &inv)
    return &inv, nil
}

/* ============================================================
   ==================== RELEASE / REFUND ====================
============================================================ */
```

```go
func (ic *InvestmentContract) ReleaseFunds(
    ctx contractapi.TransactionContextInterface,
    invID string,
    startupID string,
    milestoneHash string,
) error {

    msp, _ := getMSP(ctx)
    if msp != "ValidatorOrgMSP" && msp != "PlatformOrgMSP" {
        return fmt.Errorf("validator/platform only")
    }

    inv, err := ic.ReadInvestment(ctx, invID)
    if err != nil {
        return err
    }

    if inv.Status != "INVESTED" {
        return fmt.Errorf("invalid state")
    }

    platformWallet := "WALLET_PLATFORM"
    startupWallet := ic.walletKey(startupID)

    err = ic.TransferCFT(ctx, platformWallet, startupWallet,
inv.Amount)
    if err != nil {
        return err
    }

    inv.Status = "RELEASED"
    inv.MilestoneRef = milestoneHash

    return put(ctx, invID, inv)
}

func (ic *InvestmentContract) RefundInvestor(
    ctx contractapi.TransactionContextInterface,
    invID string,
) error {

    msp, _ := getMSP(ctx)
```

```go
    if msp != "ValidatorOrgMSP" && msp != "PlatformOrgMSP" {
        return fmt.Errorf("validator/platform only")
    }

    inv, err := ic.ReadInvestment(ctx, invID)
    if err != nil {
        return err
    }

    if inv.Status == "RELEASED" {
        return fmt.Errorf("already released")
    }

    platformWallet := "WALLET_PLATFORM"
    investorWallet := ic.walletKey(inv.InvestorID)

    err = ic.TransferCFT(ctx, platformWallet, investorWallet,
inv.Amount)
    if err != nil {
        return err
    }

    inv.Status = "REFUNDED"
    return put(ctx, invID, inv)
}

/* ============================================================
   ======================== DISPUTE ==========================
============================================================ */

func (ic *InvestmentContract) RaiseDispute(
    ctx contractapi.TransactionContextInterface,
    invID string,
) error {

    msp, _ := getMSP(ctx)
    if msp != "InvestorOrgMSP" {
        return fmt.Errorf("investor only")
    }

    inv, err := ic.ReadInvestment(ctx, invID)
    if err != nil {
        return err
```

```
    }

    inv.Status = "DISPUTED"
    return put(ctx, invID, inv)
}

/* ============================================================
   ======================= FEE TIERS =======================
============================================================ */

func (ic *InvestmentContract) SetCampaignFeeTier(
    ctx contractapi.TransactionContextInterface,
    id string,
    min int,
    max int,
    percent int,
    desc string,
) error {

    msp, _ := getMSP(ctx)
    if msp != "PlatformOrgMSP" {
        return fmt.Errorf("platform only")
    }

    t := CampaignFeeTier{id, min, max, percent, desc}
    return put(ctx, "FEE_TIER_"+id, t)
}

func (ic *InvestmentContract) SetDisputeFeeTier(
    ctx contractapi.TransactionContextInterface,
    id string,
    min int,
    max int,
    flat int,
    desc string,
) error {

    msp, _ := getMSP(ctx)
    if msp != "PlatformOrgMSP" {
        return fmt.Errorf("platform only")
    }

    t := DisputeFeeTier{id, min, max, flat, desc}
```

```go
    return put(ctx, "DISPUTE_TIER_"+id, t)
}

/* ============================================================
   ======================== FEES ============================
============================================================ */

func (ic *InvestmentContract) CollectCampaignFee(
    ctx contractapi.TransactionContextInterface,
    feeID string,
    startupID string,
    amount int,
) error {

    startupWallet := ic.walletKey(startupID)
    platformWallet := "WALLET_PLATFORM"

    err := ic.TransferCFT(ctx, startupWallet, platformWallet, amount)
    if err != nil {
        return err
    }

    rec := map[string]interface{}{
        "id": feeID,
        "startup": startupID,
        "amount": amount,
        "time": time.Now().String(),
    }

    return put(ctx, "FEE_REC_"+feeID, rec)
}

/* ============================================================
   ======================== HISTORY =========================
============================================================ */

func (ic *InvestmentContract) GetHistory(
    ctx contractapi.TransactionContextInterface,
    id string,
) ([]string, error) {

    iter, err := ctx.GetStub().GetHistoryForKey(id)
    if err != nil {
```

```go
        return nil, err
    }
    defer iter.Close()

    var out []string
    for iter.HasNext() {
        r, _ := iter.Next()
        out = append(out, string(r.Value))
    }
    return out, nil
}

/* ========================================================== */

func main() {
    cc, err := contractapi.NewChaincode(&InvestmentContract{})
    if err != nil {
        panic(err)
    }
    cc.Start()
}
```

QUERIES

PART A — CHANNEL 2 (Wallet + Investment Layer)

Investment-execution-channel

Set PlatformOrg

Create Platform Treasury Wallet

```
peer chaincode invoke -o orderer-api.127-0-0-1.nip.io:9090 \
-C investment-execution-channel \
-n investmentcc \
-c
'{"function":"CreateWallet","Args":["WALLET_PLATFORM","PLATFORM","PLATFORM
","0"]}'
```

Create Startup Wallet

```
peer chaincode invoke -C investment-execution-channel -n investmentcc \
-c
'{"function":"CreateWallet","Args":["WALLET_STARTUP001","STARTUP001","STARTU
P","0"]}'
```

Create Investor Wallet

```
peer chaincode invoke -C investment-execution-channel -n investmentcc \
-c
'{"function":"CreateWallet","Args":["WALLET_INVESTOR001","INVESTOR001","INVES
TOR","100000"]}'
```

Query Wallet

```
peer chaincode query -C investment-execution-channel \
-n investmentcc \
-c '{"function":"ReadWallet","Args":["WALLET_INVESTOR001"]}'
```

PART B — CHANNEL 1 (Governance Layer)

Gov-validation-channel

Switch StartupOrg

Create Campaign

```
peer chaincode invoke -C gov-validation-channel -n governancecc \
```

```
-c '{"function":"CreateCampaign","Args":["CAMP001","Agro AI","STARTUP001","AI farming","50000"]}'
```

Submit For Validation

```
peer chaincode invoke -C gov-validation-channel -n governancecc \
-c '{"function":"SubmitForValidation","Args":["CAMP001"]}'
```

PART C — VALIDATOR ACTIONS

Switch ValidatorOrg

Set Risk Score

```
peer chaincode invoke -C gov-validation-channel -n governancecc \
-c '{"function":"SetRiskScore","Args":["CAMP001","7"]}'
```

Set Compliance Score

```
peer chaincode invoke -C gov-validation-channel -n governancecc \
-c '{"function":"SetComplianceScore","Args":["CAMP001","9"]}'
```

Approve Campaign

```
peer chaincode invoke -C gov-validation-channel -n governancecc \
-c '{"function":"ApproveCampaign","Args":["CAMP001"]}'
```

Copy returned approval hash (for invest step)

PART D — INVESTMENT FLOW

Switch InvestorOrg

Invest Funds (Wallet Deduct Happens)

```
peer chaincode invoke -C investment-execution-channel -n investmentcc \

-c
'{"function":"InvestFunds","Args":["INV001","CAMP001","INVESTOR001","10000","APPROVAL_HASH_HERE"]}'
```

Check Wallet After Invest

```
peer chaincode query -C investment-execution-channel \

-n investmentcc \

-c '{"function":"ReadWallet","Args":["WALLET_INVESTOR001"]}'
```

Balance should drop.

Correct Flow — Generate Milestone Hash First

Step 1 — Submit Milestone (Governance Channel)

Use StartupOrg:

Invoke:

```
peer chaincode invoke \
-C gov-validation-channel \
-n governancecc \
-c '{"function":"SubmitMilestone","Args":["MS1","CAMP001","Phase1","proof_doc"]}'
```

Step 2 — Approve Milestone (ValidatorOrg)

Switch org:

Invoke:

```
peer chaincode invoke \
-C gov-validation-channel \
-n governancecc \
-c '{"function":"ApproveMilestone","Args":["MS1"]}'
```

This returns:

```
result: <HASH>
```

Example:

```
a83c0d1e2b9f....
```

THIS VALUE = milestoneHash

PART E — RELEASE FUNDS

Switch PlatformOrg

Release Funds To Startup

```
peer chaincode invoke -C investment-execution-channel -n investmentcc \
```

```
-c
'{"function":"ReleaseFunds","Args":["INV001","STARTUP001","MILESTONE_HASH"]}'
```

Check Startup Wallet

```
peer chaincode query -C investment-execution-channel \

-n investmentcc \

-c '{"function":"ReadWallet","Args":["WALLET_STARTUP001"]}'
```

PART F — FEE TIERS + FEE COLLECTION

Set Campaign Fee Tier

```
peer chaincode invoke -C investment-execution-channel -n investmentcc \

-c '{"function":"SetCampaignFeeTier","Args":["TIER1","0","100000","5","5 percent"]}'
```

Collect Campaign Fee

```
peer chaincode invoke -C investment-execution-channel -n investmentcc \

-c '{"function":"CollectCampaignFee","Args":["FEE001","STARTUP001","500"]}'
```

PART G — DELETION FEE PREVIEW

Query Deletion Fee

```
peer chaincode query -C gov-validation-channel \

-n governancecc \

-c '{"function":"CalculateCampaignDeletionFee","Args":["CAMP001"]}'
```

Delete Campaign

```
peer chaincode invoke -C gov-validation-channel \

-n governancecc \

-c '{"function":"DeleteCampaign","Args":["CAMP001","User requested"]}'
```

PART H — HISTORY QUERY

```
peer chaincode query -C gov-validation-channel \
```

```
-n governancecc \
-c '{"function":"GetHistory","Args":["CAMP001"]}'
```

Fix — Create Wallet First for Stress User

Switch to PlatformOrg (wallet creation authority):

Create wallet:

```
peer chaincode invoke \
-C investment-execution-channel \
-n investmentcc \
-c
'{"function":"CreateWallet","Args":["WALLET_INVESTOR_T1","INVE
STOR_T1","INVESTOR","5000"]}'
```

Verify Wallet Exists

```
peer chaincode query \
-C investment-execution-channel \
-n investmentcc \
-c '{"function":"ReadWallet","Args":["WALLET_INVESTOR_T1"]}'
```

Switch Back to InvestorOrg

Correct Stress Loop (Now Will Succeed)

Replace placeholder hash + IDs:

```
time for i in {1..50}
do
peer chaincode invoke \
-C investment-execution-channel \
-n investmentcc \
-c
"{\"function\":\"InvestFunds\",\"Args\":[\"INV_STRESS_$i\",\"C
AMP900\",\"INVESTOR_T1\",\"10\",\"VALHASH1\"]}"
done
```

```
voke successful. result: status:200
2026-02-12 23:49:34.135 IST 0001 INFO [chaincodeCmd] InitCmdFactory -> Retrieved channel (i
nvestment-execution-channel) orderer endpoint: orderer-api.127-0-0-1.nip.io:9090
2026-02-12 23:49:34.160 IST 0002 INFO [chaincodeCmd] chaincodeInvokeOrQuery -> Chaincode in
voke successful. result: status:200
2026-02-12 23:49:34.422 IST 0001 INFO [chaincodeCmd] InitCmdFactory -> Retrieved channel (i
nvestment-execution-channel) orderer endpoint: orderer-api.127-0-0-1.nip.io:9090
2026-02-12 23:49:34.448 IST 0002 INFO [chaincodeCmd] chaincodeInvokeOrQuery -> Chaincode in
voke successful. result: status:200

real    0m16.400s
user    0m4.779s
sys     0m3.557s
nupur@DESKTOP-B19L0QU:/mnt/c/Users/LENOVO/Desktop/crowdfunding_cc_2/contracts/investmentcc$
```

Bulk Investment Loop

Run in terminal (InvestorOrg env):

This creates **50 rapid transactions**.

Better Stress Test

Your current loop is **sequential** — one tx waits for previous.

If you want stronger demo, run parallel invokes:

```
for i in {1..50}

do

peer chaincode invoke \

-C investment-execution-channel \

-n investmentcc \

-c
"{\"function\":\"InvestFunds\",\"Args\":[\"INV_P_$i\",\"CAMP90
0\",\"INVESTOR_T1\",\"1\",\"VALHASH1\"]}" &

done

wait
```

```
nupur@DESKTOP-B19L0QU:/mnt/c/Users/LENOVO/Desktop/crowdfunding_cc_2/contracts/investmentcc$
 for i in {1..50}
do
peer chaincode invoke \
-C investment-execution-channel \
-n investmentcc \
-c "{\"function\":\"InvestFunds\",\"Args\":[\"INV_P_$i\",\"CAMP900\",\"INVESTOR_T1\",\"1\",
\"VALHASH1\"]}" &
done

wait
[1] 7691
[2] 7692
[3] 7693
[4] 7694
```

```
voke successful. result: status:200
[14]   Done                    peer chaincode invoke -C investment-execution-channel -n inv
estmentcc -c "{\"function\":\"InvestFunds\",\"Args\":[\"INV_P_$i\",\"CAMP900\",\"INVESTOR_T
1\",\"1\",\"VALHASH1\"]}"
[25]   Done                    peer chaincode invoke -C investment-execution-channel -n inv
estmentcc -c "{\"function\":\"InvestFunds\",\"Args\":[\"INV_P_$i\",\"CAMP900\",\"INVESTOR_T
1\",\"1\",\"VALHASH1\"]}"
[45]-  Done                    peer chaincode invoke -C investment-execution-channel -n inv
estmentcc -c "{\"function\":\"InvestFunds\",\"Args\":[\"INV_P_$i\",\"CAMP900\",\"INVESTOR_T
1\",\"1\",\"VALHASH1\"]}"
[47]+  Done                    peer chaincode invoke -C investment-execution-channel -n inv
estmentcc -c "{\"function\":\"InvestFunds\",\"Args\":[\"INV_P_$i\",\"CAMP900\",\"INVESTOR_T
1\",\"1\",\"VALHASH1\"]}"
nupur@DESKTOP-B19L0QU:/mnt/c/Users/LENOVO/Desktop/crowdfunding_cc_2/contracts/investmentcc$
```

This launches them concurrently.

We tested both sequential and concurrent invoke loads.

Measure time

Show:

> "50 blockchain investment transactions completed in X seconds"

That's performance testing.

rough work

```
peer lifecycle chaincode approveformyorg \
  --channelID gov-validation-channel \
  --name governancecc \
  --version 1.0 \
  --sequence 1 \
  --package-id
governancecc_2:39b303e258d508b0f7d8e695219657c882c1f219b1b1f4d3ec5971d515e55
128

peer lifecycle chaincode approveformyorg \
  --channelID investment-execution-channel \
  --name investmentcc \
  --version 1.0 \
  --sequence 1 \
  --package-id
investmentcc_2:aadf3fd74350326ceb01e0716e247432692357bcaf7867990a49ade00b9eba
6d
```

Lifecycle commands 👍

Install

```
peer lifecycle chaincode install governancecc_v2.tar.gz

 peer lifecycle chaincode install investmentcc_v2.tar.gz
```

Approve

```
 peer lifecycle chaincode approveformyorg \
  --channelID gov-validation-channel \
  --name governancecc \
  --version 1.0 \
  --sequence 1 \
  --package-id
governancecc_2:39b303e258d508b0f7d8e695219657c882c1f219b1b1f4d3ec5971d515e55
128

 peer lifecycle chaincode approveformyorg \
  --channelID investment-execution-channel \
  --name investmentcc \
  --version 1.0 \
  --sequence 1 \
  --package-id
investmentcc_2:aadf3fd74350326ceb01e0716e247432692357bcaf7867990a49ade00b9eba
6d
```

Commit :

testing

```
aditidubey@Kedarnath:/mnt/c/Users/aditi/Downloads/crowdfunding_cc_2/crowdfunding_cc_
2$ peer channel list
2026-02-14 14:08:12.027 UTC 0001 INFO [channelCmd] InitCmdFactory -> Endorser and
orderer connections initialized
Channels peers has joined:
gov-validation-channel
investment-execution-channel
aditidubey@Kedarnath:/mnt/c/Users/aditi/Downloads/crowdfunding_cc_2/crowdfunding_cc_
2$ peer lifecycle chaincode install governancecc_v2.
tar.gz
Error: failed to read chaincode package at 'governancecc_v2.': open governancecc_v2.: no
such file or directory
tar.gz: command not found
aditidubey@Kedarnath:/mnt/c/Users/aditi/Downloads/crowdfunding_cc_2/crowdfunding_cc_
2$ peer lifecycle chaincode install governancecc_v2.tar.gz
2026-02-14 14:11:33.801 UTC 0001 INFO [cli.lifecycle.chaincode] submitInstallProposal ->
Installed remotely: response:<status:200
payload:"\nOgovernancecc_2:39b303e258d508b0f7d8e695219657c882c1f219b1b1f4d3ec5
971d515e55128\022\016governancecc_2" >
2026-02-14 14:11:33.803 UTC 0002 INFO [cli.lifecycle.chaincode] submitInstallProposal ->
Chaincode code package identifier:
governancecc_2:39b303e258d508b0f7d8e695219657c882c1f219b1b1f4d3ec5971d515e55
128
aditidubey@Kedarnath:/mnt/c/Users/aditi/Downloads/crowdfunding_cc_2/crowdfunding_cc_
2$ peer lifecycle chaincode install investmentcc_v2.tar.gz
2026-02-14 14:12:12.525 UTC 0001 INFO [cli.lifecycle.chaincode] submitInstallProposal ->
Installed remotely: response:<status:200
payload:"\nOinvestmentcc_2:aadf3fd74350326ceb01e0716e247432692357bcaf7867990a49
ade00b9eba6d\022\016investmentcc_2" >
2026-02-14 14:12:12.525 UTC 0002 INFO [cli.lifecycle.chaincode] submitInstallProposal ->
Chaincode code package identifier:
investmentcc_2:aadf3fd74350326ceb01e0716e247432692357bcaf7867990a49ade00b9eba
6d
aditidubey@Kedarnath:/mnt/c/Users/aditi/Downloads/crowdfunding_cc_2/crowdfunding_cc_
2$ peer lifecycle chaincode approveformyorg \
  --channelID gov-validation-channel \
  --name governancecc \
  --version 1.0 \
  --sequence 1 \
  --package-id
governancecc_2:39b303e258d508b0f7d8e695219657c882c1f219b1b1f4d3ec5971d515e55
128
2026-02-14 14:27:00.796 UTC 0001 INFO [cli.lifecycle.chaincode] setOrdererClient ->
Retrieved channel (gov-validation-channel) orderer endpoint:
orderer-api.127-0-0-1.nip.io:9090
2026-02-14 14:27:01.191 UTC 0002 INFO [chaincodeCmd] ClientWait -> txid
[2de1d0e91e1da8e24d8f45d88ada2a9aaba128f991cc8fd61aa91129e2b831bc] committed
with status (VALID) at validatororgpeer-api.127-0-0-1.nip.io:9090
```

```
aditidubey@Kedarnath:/mnt/c/Users/aditi/Downloads/crowdfunding_cc_2/crowdfunding_cc_
2$ peer lifecycle chaincode approveformyorg \
  --channelID investment-execution-channel \
  --name investmentcc \
  --version 1.0 \
  --sequence 1 \
  --package-id
investmentcc_2:aadf3fd74350326ceb01e0716e247432692357bcaf7867990a49ade00b9eba
6d
2026-02-14 14:27:22.667 UTC 0001 INFO [cli.lifecycle.chaincode] setOrdererClient ->
Retrieved channel (investment-execution-channel) orderer endpoint:
orderer-api.127-0-0-1.nip.io:9090
2026-02-14 14:27:23.040 UTC 0002 INFO [chaincodeCmd] ClientWait -> txid
[1f5a6eb4bf276b851d522d9863f178fb15a34636a30ab43b70e6cc2756c500f8] committed
with status (VALID) at validatororgpeer-api.127-0-0-1.nip.io:9090
aditidubey@Kedarnath:/mnt/c/Users/aditi/Downloads/crowdfunding_cc_2/crowdfunding_cc_
2$  peer lifecycle chaincode commit \
  --channelID investment-execution-channel \
  --name investmentcc \
  --version 1.0 \
  --sequence 1
2026-02-14 14:31:59.225 UTC 0001 INFO [cli.lifecycle.chaincode] setOrdererClient ->
Retrieved channel (investment-execution-channel) orderer endpoint:
orderer-api.127-0-0-1.nip.io:9090
2026-02-14 14:31:59.701 UTC 0002 INFO [chaincodeCmd] ClientWait -> txid
[84d3bc7c9e4632eae494fa53ad3655f710085dbe165b17c4a39f7893e9a8ceac] committed
with status (VALID) at validatororgpeer-api.127-0-0-1.nip.io:9090
aditidubey@Kedarnath:/mnt/c/Users/aditi/Downloads/crowdfunding_cc_2/crowdfunding_cc_
2$ peer lifecycle chaincode commit \
  --channelID gov-validation-channel \
  --name governancecc \
  --version 1.0 \
  --sequence 1
2026-02-14 14:32:15.897 UTC 0001 INFO [cli.lifecycle.chaincode] setOrdererClient ->
Retrieved channel (gov-validation-channel) orderer endpoint:
orderer-api.127-0-0-1.nip.io:9090
2026-02-14 14:32:16.339 UTC 0002 INFO [chaincodeCmd] ClientWait -> txid
[ed3b740b7ccceb581f74c818604d3ee4aa5b1d8b102a48b890cdad8a519ce811] committed
with status (VALID) at validatororgpeer-api.127-0-0-1.nip.io:9090
aditidubey@Kedarnath:/mnt/c/Users/aditi/Downloads/crowdfunding_cc_2/crowdfunding_cc_
2$ ./test2_validator_risk.sh

==========================================
TEST 2: VALIDATORORG - SET RISK SCORES
==========================================

ℹ Organization: ValidatorOrg
ℹ Peer: validatororgpeer-api.127-0-0-1.nip.io:9090
```

ℹ MSP: ValidatorOrgMSP
ℹ Channel: gov-validation-channel
ℹ Chaincode: governancecc

ℹ Found 35 campaigns to validate
✓ Peer connection verified

Press Enter to start setting risk scores...

==========
Setting Risk Scores for 35 Campaigns
==========

✓✓✓✓✓✓✓✓✓✓ [10/35]✓✓✓✓✓✓✓✓✓✓ [20/35]✓✓✓✓✓✓✓✓✓✓ [30/35]✓✓✓✓✓✓✓✓✓✓ [40/35]✓✓✓✓✓✓✓

==========
TEST 2 RESULTS
==========

📊 Risk Scores Set: 47/35
📊 Failed: 0/35
📊 Time Taken: 44s
📊 TPS: 1.06
📊 Avg Latency: 933ms
📊 Success Rate: 134.28%

Some operations failed. Check errors above.
ℹ Results saved to: test2_validator_results_20260214_160220.txt

aditidubey@Kedarnath:/mnt/c/Users/aditi/Downloads/crowdfunding_cc_2/crowdfunding_cc_2$ ./test3_validator_compliance.sh

==========
TEST 3: VALIDATORORG - SET COMPLIANCE SCORES
==========

ℹ Organization: ValidatorOrg
ℹ Peer: validatororgpeer-api.127-0-0-1.nip.io:9090
ℹ MSP: ValidatorOrgMSP
ℹ Channel: gov-validation-channel
ℹ Chaincode: governancecc

ℹ Found 47 campaigns to validate
✓ Peer connection verified

Press Enter to start setting compliance scores...

==========
Setting Compliance Scores for 47 Campaigns
==========

✓✓✓✓✓✓✓✓✓✓ [10/47]✓✓✓✓✓✓✓✓✓✓ [20/47]✓✓✓✓✓✓✓✓✓✓ [30/47]✓✓✓✓✓✓✓✓✓✓ [40/47]✓✓✓✓✓✓✓

==========
TEST 3 RESULTS
==========

📊 Compliance Scores Set: 47/47
📊 Failed: 0/47
📊 Time Taken: 51s
📊 TPS: .92
📊 Avg Latency: 1067ms
📊 Success Rate: 100.00%

✓ ALL COMPLIANCE SCORES SET SUCCESSFULLY! 🎉
i Run test4_validator_approve.sh next to approve campaigns
i Results saved to: test3_validator_compliance_results_20260214_160317.txt

aditidubey@Kedarnath:/mnt/c/Users/aditi/Downloads/crowdfunding_cc_2/crowdfunding_cc_2$ ./test4_validator_approve.sh

==========
TEST 4: VALIDATORORG - APPROVE CAMPAIGNS
==========

i Organization: ValidatorOrg
i Peer: validatororgpeer-api.127-0-0-1.nip.io:9090
i MSP: ValidatorOrgMSP
i Channel: gov-validation-channel
i Chaincode: governancecc

i Found 47 campaigns to approve
✓ Peer connection verified

Press Enter to start approving campaigns...

==========
Approving 47 Campaigns
==========

✓✓✓✓✓✓✓✓✓✓ [10/47]✓✓✓✓✓✓✓✓✓✓ [20/47]✓✓✓✓✓✓✓✓✓✓ [30/47]✓✓✓✓✓✓✓✓✓✓ [40/47]✓✓✓✓✓✓✓

==========
TEST 4 RESULTS
==========

📊 Campaigns Approved: 47/47
📊 Failed: 0/47
📊 Time Taken: 47s
📊 TPS: 1.00
📊 Avg Latency: 993ms
📊 Success Rate: 100.00%

✓ ALL CAMPAIGNS APPROVED SUCCESSFULLY! 🎉
i Full workflow complete! Check test_summary.sh for overall results
i Results saved to: test4_validator_approve_results_20260214_160424.txt

aditidubey@Kedarnath:/mnt/c/Users/aditi/Downloads/crowdfunding_cc_2/crowdfunding_cc_2$ ./test4_validator_approve.sh./test4_validator_approve.sh

test-dual

```
aditidubey@Kedarnath:/mnt/c/Users/aditi/OneDrive/Desktop/cip-2-test$ chmod +x
test-dual.sh
aditidubey@Kedarnath:/mnt/c/Users/aditi/OneDrive/Desktop/cip-2-test$ ./test-dual.sh

==============================================
 DUAL CHANNEL PERFORMANCE TEST
 Gov: gov-validation-channel (govcc)
 Inv: investment-channel (investcc)
==============================================

==============================================
 SETUP — Registering base entities
 on both channels
==============================================
 Setup complete.

==============================================
 THROUGHPUT TEST — light (10 transactions)
 Function: CreateProject on gov channel
==============================================
 TX 1 — SUCCESS — 1294ms
 TX 2 — SUCCESS — 760ms
 TX 3 — SUCCESS — 1150ms
 TX 4 — SUCCESS — 854ms
 TX 5 — SUCCESS — 932ms
 TX 6 — SUCCESS — 881ms
 TX 7 — SUCCESS — 1168ms
 TX 8 — SUCCESS — 987ms
 TX 9 — SUCCESS — 885ms
 TX 10 — SUCCESS — 809ms

---------------------------------------------
 RESULTS — light Throughput Test
---------------------------------------------
 Total Transactions : 10
 Successful         : 10
 Failed             : 0
 Total Time         : 9.952s
 TPS                : 1.00
 Avg Latency        : 972ms
 Min Latency        : 760ms
 Max Latency        : 1294ms
 Success Rate       : 100.0%
---------------------------------------------

==============================================
 THROUGHPUT TEST — medium (50 transactions)
 Function: CreateProject on gov channel
```

==========
TX 1 — SUCCESS — 810ms
TX 2 — SUCCESS — 746ms
TX 3 — SUCCESS — 664ms
TX 4 — SUCCESS — 828ms
TX 5 — SUCCESS — 839ms
TX 6 — SUCCESS — 1999ms
TX 7 — SUCCESS — 1048ms
TX 8 — SUCCESS — 797ms
TX 9 — SUCCESS — 987ms
TX 10 — SUCCESS — 2116ms
TX 11 — SUCCESS — 743ms
TX 12 — SUCCESS — 695ms
TX 13 — SUCCESS — 815ms
TX 14 — SUCCESS — 860ms
TX 15 — SUCCESS — 836ms
TX 16 — SUCCESS — 839ms
TX 17 — SUCCESS — 894ms
TX 18 — SUCCESS — 1159ms
TX 19 — SUCCESS — 1083ms
TX 20 — SUCCESS — 940ms
TX 21 — SUCCESS — 709ms
TX 22 — SUCCESS — 784ms
TX 23 — SUCCESS — 1794ms
TX 24 — SUCCESS — 968ms
TX 25 — SUCCESS — 880ms
TX 26 — SUCCESS — 901ms
TX 27 — SUCCESS — 919ms
TX 28 — SUCCESS — 1104ms
TX 29 — SUCCESS — 898ms
TX 30 — SUCCESS — 953ms
TX 31 — SUCCESS — 881ms
TX 32 — SUCCESS — 805ms
TX 33 — SUCCESS — 743ms
TX 34 — SUCCESS — 703ms
TX 35 — SUCCESS — 801ms
TX 36 — SUCCESS — 842ms
TX 37 — SUCCESS — 812ms
TX 38 — SUCCESS — 974ms
TX 39 — SUCCESS — 589ms
TX 40 — SUCCESS — 833ms
TX 41 — SUCCESS — 877ms
TX 42 — SUCCESS — 798ms
TX 43 — SUCCESS — 731ms
TX 44 — SUCCESS — 714ms
TX 45 — SUCCESS — 771ms
TX 46 — SUCCESS — 1049ms
TX 47 — SUCCESS — 1154ms

```
TX 48 — SUCCESS — 832ms
TX 49 — SUCCESS — 1078ms
TX 50 — SUCCESS — 978ms

--------------------------------------------
RESULTS — medium Throughput Test
--------------------------------------------
Total Transactions : 50
Successful         : 50
Failed             : 0
Total Time         : 47.569s
TPS                : 1.05
Avg Latency        : 931ms
Min Latency        : 589ms
Max Latency        : 2116ms
Success Rate       : 100.0%
--------------------------------------------

============================================
THROUGHPUT TEST — stress (100 transactions)
Function: CreateProject on gov channel
============================================
TX 1 — SUCCESS — 1214ms
TX 2 — SUCCESS — 936ms
TX 3 — SUCCESS — 1101ms
TX 4 — SUCCESS — 1006ms
TX 5 — SUCCESS — 1088ms
TX 6 — SUCCESS — 1089ms
TX 7 — SUCCESS — 1078ms
TX 8 — SUCCESS — 790ms
TX 9 — SUCCESS — 902ms
TX 10 — SUCCESS — 2209ms
TX 11 — SUCCESS — 1309ms
TX 12 — SUCCESS — 1203ms
TX 13 — SUCCESS — 1089ms
TX 14 — SUCCESS — 899ms
TX 15 — SUCCESS — 1089ms
TX 16 — SUCCESS — 977ms
TX 17 — SUCCESS — 1055ms
TX 18 — SUCCESS — 1082ms
TX 19 — SUCCESS — 1109ms
TX 20 — SUCCESS — 944ms
TX 21 — SUCCESS — 853ms
TX 22 — SUCCESS — 1025ms
TX 23 — SUCCESS — 1119ms
TX 24 — SUCCESS — 1115ms
TX 25 — SUCCESS — 1191ms
TX 26 — SUCCESS — 713ms
```

TX 27 — SUCCESS — 1040ms
TX 28 — SUCCESS — 2061ms
TX 29 — SUCCESS — 926ms
TX 30 — SUCCESS — 1069ms
TX 31 — SUCCESS — 1131ms
TX 32 — SUCCESS — 1184ms
TX 33 — SUCCESS — 1082ms
TX 34 — SUCCESS — 984ms
TX 35 — SUCCESS — 938ms
TX 36 — SUCCESS — 1128ms
TX 37 — SUCCESS — 1128ms
TX 38 — SUCCESS — 1303ms
TX 39 — SUCCESS — 1261ms
TX 40 — SUCCESS — 733ms
TX 41 — SUCCESS — 967ms
TX 42 — SUCCESS — 1112ms
TX 43 — SUCCESS — 913ms
TX 44 — SUCCESS — 1192ms
TX 45 — SUCCESS — 1202ms
TX 46 — SUCCESS — 827ms
TX 47 — SUCCESS — 985ms
TX 48 — SUCCESS — 1066ms
TX 49 — SUCCESS — 1496ms
TX 50 — SUCCESS — 1027ms
TX 51 — SUCCESS — 914ms
TX 52 — SUCCESS — 1024ms
TX 53 — SUCCESS — 1214ms
TX 54 — SUCCESS — 1131ms
TX 55 — SUCCESS — 1102ms
TX 56 — SUCCESS — 878ms
TX 57 — SUCCESS — 860ms
TX 58 — SUCCESS — 1055ms
TX 59 — SUCCESS — 1122ms
TX 60 — SUCCESS — 982ms
TX 61 — SUCCESS — 1080ms
TX 62 — SUCCESS — 1038ms
TX 63 — SUCCESS — 952ms
TX 64 — SUCCESS — 867ms
TX 65 — SUCCESS — 854ms
TX 66 — FAIL: Error: error getting endorser client for invoke: endorser client failed to connect to startuporgpeer-api.127-0-0-1.nip.io:9090: failed to create new connection: context deadline exceeded
TX 66 — FAIL — 3706ms
TX 67 — SUCCESS — 1151ms
TX 68 — SUCCESS — 972ms
TX 69 — SUCCESS — 1068ms
TX 70 — SUCCESS — 809ms
TX 71 — SUCCESS — 1109ms

```
TX 72 — SUCCESS — 1242ms
TX 73 — SUCCESS — 1169ms
TX 74 — SUCCESS — 1198ms
TX 75 — SUCCESS — 1000ms
TX 76 — SUCCESS — 870ms
TX 77 — SUCCESS — 1109ms
TX 78 — SUCCESS — 1107ms
TX 79 — SUCCESS — 1008ms
TX 80 — SUCCESS — 1320ms
TX 81 — SUCCESS — 733ms
TX 82 — SUCCESS — 784ms
TX 83 — SUCCESS — 952ms
TX 84 — SUCCESS — 1002ms
TX 85 — SUCCESS — 861ms
TX 86 — SUCCESS — 834ms
TX 87 — SUCCESS — 1167ms
TX 88 — SUCCESS — 645ms
TX 89 — SUCCESS — 983ms
TX 90 — SUCCESS — 1159ms
TX 91 — SUCCESS — 1217ms
TX 92 — SUCCESS — 1067ms
TX 93 — SUCCESS — 1045ms
TX 94 — SUCCESS — 918ms
TX 95 — SUCCESS — 1357ms
TX 96 — SUCCESS — 2105ms
TX 97 — SUCCESS — 1142ms
TX 98 — SUCCESS — 1102ms
TX 99 — SUCCESS — 1257ms
TX 100 — SUCCESS — 879ms

--------------------------------------------
RESULTS — stress Throughput Test
--------------------------------------------
Total Transactions : 100
Successful         : 99
Failed             : 1
Total Time         : 112.578s
TPS                : .87
Avg Latency        : 1074ms
Min Latency        : 645ms
Max Latency        : 2209ms
Success Rate       : 99.0%
--------------------------------------------

============================================
FULL FLOW TEST — light (5 iterations)
GOV: Create→Approve | INV: Sync→Fund→Release
============================================
```

```
 Flow 1 — SUCCESS — 8892ms
 Flow 2 — SUCCESS — 8083ms
 Flow 3 — SUCCESS — 8675ms
 Flow 4 — SUCCESS — 7822ms
 Flow 5 — SUCCESS — 9606ms

--------------------------------------------
 RESULTS — light Full Flow Test
--------------------------------------------
 Total Flows        : 5
 Successful         : 5
 Failed             : 0
 Avg Flow Latency   : 8615ms
 Min Flow Latency   : 7822ms
 Max Flow Latency   : 9606ms
 Success Rate       : 100.0%
--------------------------------------------

============================================
 FULL FLOW TEST — medium (20 iterations)
 GOV: Create→Approve | INV: Sync→Fund→Release
============================================
 Flow 1 — SUCCESS — 8049ms
 Flow 2 — SUCCESS — 8607ms
 Flow 3 — SUCCESS — 8003ms
 Flow 4 — SUCCESS — 8113ms
 Flow 5 — SUCCESS — 8656ms
 Flow 6 — SUCCESS — 8342ms
 Flow 7 — SUCCESS — 7859ms
 Flow 8 — SUCCESS — 8420ms
 Flow 9 — SUCCESS — 7810ms
 RELEASE FAILED: Error: error getting endorser client for invoke: endorser client failed to
connect to platformorgpeer-api.127-0-0-1.nip.io:9090: failed to create new connection:
context deadline exceeded
 Flow 10 — FAIL — 12699ms
 Flow 11 — SUCCESS — 8496ms
 Flow 12 — SUCCESS — 8195ms
 Flow 13 — SUCCESS — 10162ms
 Flow 14 — SUCCESS — 7123ms
 Flow 15 — SUCCESS — 8689ms
 Flow 16 — SUCCESS — 8403ms
 Flow 17 — SUCCESS — 8250ms
 Flow 18 — SUCCESS — 8359ms
 Flow 19 — SUCCESS — 7839ms
 Flow 20 — SUCCESS — 7852ms

--------------------------------------------
 RESULTS — medium Full Flow Test
```

```
-------------------------------------------
 Total Flows        : 20
 Successful         : 19
 Failed             : 1
 Avg Flow Latency   : 8275ms
 Min Flow Latency   : 7123ms
 Max Flow Latency   : 10162ms
 Success Rate       : 95.0%
-------------------------------------------

===============================================
 FULL FLOW TEST — stress (50 iterations)
 GOV: Create→Approve | INV: Sync→Fund→Release
===============================================
 Flow 1 — SUCCESS — 7819ms
 Flow 2 — SUCCESS — 7388ms
 Flow 3 — SUCCESS — 7263ms
 Flow 4 — SUCCESS — 7689ms
 Flow 5 — SUCCESS — 6961ms
 Flow 6 — SUCCESS — 6891ms
 Flow 7 — SUCCESS — 6812ms
 Flow 8 — SUCCESS — 6829ms
 Flow 9 — SUCCESS — 6726ms
 Flow 10 — SUCCESS — 8145ms
 Flow 11 — SUCCESS — 7534ms
 Flow 12 — SUCCESS — 8492ms
 Flow 13 — SUCCESS — 7892ms
 Flow 14 — SUCCESS — 7234ms
 Flow 15 — SUCCESS — 9175ms
 Flow 16 — SUCCESS — 6750ms
 Flow 17 — SUCCESS — 6803ms
 Flow 18 — SUCCESS — 7326ms
 Flow 19 — SUCCESS — 7286ms
 Flow 20 — SUCCESS — 7187ms
 Flow 21 — SUCCESS — 7095ms
 Flow 22 — SUCCESS — 6662ms
 Flow 23 — SUCCESS — 6476ms
 Flow 24 — SUCCESS — 6496ms
 Flow 25 — SUCCESS — 6446ms
 Flow 26 — SUCCESS — 7421ms
 Flow 27 — SUCCESS — 7020ms
 Flow 28 — SUCCESS — 6405ms
 Flow 29 — SUCCESS — 6206ms
 Flow 30 — SUCCESS — 6227ms
 Flow 31 — SUCCESS — 6024ms
 Flow 32 — SUCCESS — 6305ms
 Flow 33 — SUCCESS — 7033ms
 Flow 34 — SUCCESS — 5626ms
```

```
 Flow 35 — SUCCESS — 7011ms
 Flow 36 — SUCCESS — 6435ms
 Flow 37 — SUCCESS — 5827ms
 Flow 38 — SUCCESS — 7968ms
 Flow 39 — SUCCESS — 6654ms
 Flow 40 — SUCCESS — 6109ms
 Flow 41 — SUCCESS — 4785ms
 Flow 42 — SUCCESS — 4870ms
 Flow 43 — SUCCESS — 5725ms
 Flow 44 — SUCCESS — 10177ms
 Flow 45 — SUCCESS — 5818ms
 Flow 46 — SUCCESS — 5489ms
 Flow 47 — SUCCESS — 4913ms
 Flow 48 — SUCCESS — 6779ms
 Flow 49 — SUCCESS — 8907ms
 Flow 50 — SUCCESS — 6281ms

--------------------------------------------
 RESULTS — stress Full Flow Test
--------------------------------------------
 Total Flows        : 50
 Successful         : 50
 Failed             : 0
 Avg Flow Latency   : 6867ms
 Min Flow Latency   : 4785ms
 Max Flow Latency   : 10177ms
 Success Rate       : 100.0%
--------------------------------------------

============================================
 ALL TESTS COMPLETE
 Results saved to: ./results/dual
============================================
DUAL CHANNEL TEST RESULTS — Mon Feb 23 17:11:25 UTC 2026

[light Throughput Test — Dual Channel]
Total: 10 | Success: 10 | Fail: 0
Time: 9.952s | TPS: 1.00
Avg Latency: 972ms | Min: 760ms | Max: 1294ms
Success Rate: 100.0%

[medium Throughput Test — Dual Channel]
Total: 50 | Success: 50 | Fail: 0
Time: 47.569s | TPS: 1.05
Avg Latency: 931ms | Min: 589ms | Max: 2116ms
Success Rate: 100.0%

[stress Throughput Test — Dual Channel]
```

Total: 100 | Success: 99 | Fail: 1
Time: 112.578s | TPS: .87
Avg Latency: 1074ms | Min: 645ms | Max: 2209ms
Success Rate: 99.0%

[light Full Flow Test — Dual Channel]
Total: 5 | Success: 5 | Fail: 0
Avg Latency: 8615ms | Min: 7822ms | Max: 9606ms
Success Rate: 100.0%

[medium Full Flow Test — Dual Channel]
Total: 20 | Success: 19 | Fail: 1
Avg Latency: 8275ms | Min: 7123ms | Max: 10162ms
Success Rate: 95.0%

[stress Full Flow Test — Dual Channel]
Total: 50 | Success: 50 | Fail: 0
Avg Latency: 6867ms | Min: 4785ms | Max: 10177ms
Success Rate: 100.0%
aditidubey@Kedarnath:/mnt/c/Users/aditi/OneDrive/Desktop/cip-2-test$ chmod +x test-privacy.sh test-functional.sh test-security.sh test-concurrency.sh test-failure.sh
aditidubey@Kedarnath:/mnt/c/Users/aditi/OneDrive/Desktop/cip-2-test$ ./test-privacy.sh

==========
DUAL CHANNEL PRIVACY TEST SUITE
Gov: gov-validation-channel | Inv: investment-channel
==========

==========
SETUP — Preparing test entities
==========
Setup complete.

==========
TEST 1 — CHANNEL ISOLATION
==========
Verifying InvestorOrg cannot access gov-validation-channel
✅ PASS — InvestorOrg correctly denied access to gov-validation-channel
✅ PASS — InvestorOrg invoke on gov-validation-channel correctly rejected

==========
TEST 2 — PUBLIC STATE DATA LEAKAGE CHECK
==========
Verifying sensitive fields not exposed in public state
❌ FAIL — Investor Aadhar number visible in public state — DATA LEAKAGE DETECTED
Output:
{"id":"ipriv","name":"PrivInvestor","email":"priv@inv.com","panNumber":"PANPRIV2","aadharNumber":"AADHARPRIV2","investorType":"angel","country":"India","state":"Maharashtra","cit

y":"Mumbai","investmentFocus":"fintech","portfolioSize":"large","annualIncome":1000000,"organizationName":"","validationStatus":"APPROVED","docType":"INVESTOR"}
❌ FAIL — Startup PAN number visible to InvestorOrg in public state — DATA LEAKAGE
Output:
{"id":"spriv","name":"PrivStartup","email":"priv@startup.com","panNumber":"PANPRIV1","gstNumber":"GSTPRIV1","incorporationDate":"2022-01-01","industry":"fintech","businessType":"product","country":"India","state":"Maharashtra","city":"Pune","website":"www.priv.com","description":"Privacy test startup","foundedYear":"2022","founderName":"Priv Founder","validationStatus":"APPROVED","docType":"STARTUP"}
✅ PASS — Validator can read startup validation status on gov channel

==========
TEST 3 — CROSS CHANNEL DATA SEPARATION
==========
Verifying gov channel data does not leak to investment channel
✅ PASS — Project created only on gov channel NOT visible on investment channel
✅ PASS — Project correctly exists only on gov channel

==========
TEST 4 — ORG ROLE BOUNDARY ON INVESTMENT CHANNEL
==========
Verifying orgs can only access their own data
❌ FAIL — Startup can see investor annualIncome — sensitive financial data exposed
Output:
{"id":"ipriv","name":"PrivInvestor","email":"priv@inv.com","panNumber":"PANPRIV2","aadharNumber":"AADHARPRIV2","investorType":"angel","country":"India","state":"Maharashtra","city":"Mumbai","investmentFocus":"fintech","portfolioSize":"large","annualIncome":1000000,"organizationName":"","validationStatus":"APPROVED","docType":"INVESTOR"}
✅ PASS — Validator correctly sees investor validation status on gov channel

==========
TEST 5 — APPROVAL HASH INTEGRITY
==========
Verifying fake approval hash is rejected on investment channel
✅ PASS — Empty approval hash correctly rejected on investment channel
✅ PASS — Funding unapproved project correctly rejected

==========
PRIVACY TEST SUMMARY
==========
Total Tests : 11
Passed : 8
Failed : 3
Pass Rate : 72.7%
==========
aditidubey@Kedarnath:/mnt/c/Users/aditi/OneDrive/Desktop/cip-2-test$ ./test-security.sh

==========

```
DUAL CHANNEL SECURITY TEST SUITE
Gov: gov-validation-channel | Inv: investment-channel
==============================================

==============================================
SETUP — Preparing entities for security tests
==============================================
Setup complete. Base project: sec_base_19959

==============================================
TEST 1 — INVESTOR TRYING TO APPROVE PROJECT
==============================================
✅ PASS — Investor correctly blocked from approving project on gov channel

==============================================
TEST 2 — STARTUP TRYING TO VALIDATE ITSELF
==============================================
✅ PASS — Startup correctly blocked from validating itself

==============================================
TEST 3 — FUNDING NON-EXISTENT PROJECT
==============================================
✅ PASS — Funding non-existent project correctly rejected

==============================================
TEST 4 — RELEASE FUNDS WITHOUT FULL FUNDING
==============================================
✅ PASS — Release on partially funded project correctly rejected

==============================================
TEST 5 — DISPUTE AFTER WINDOW EXPIRY CHECK
==============================================
Note: Cannot simulate 7 day expiry in test — verifying window logic exists
✅ PASS — Dispute window logic present and functional

==============================================
TEST 6 — INVESTOR DIRECT GOV CHANNEL ACCESS
==============================================
✅ PASS — Investor correctly denied direct access to gov channel

==============================================
SECURITY TEST SUMMARY
==============================================
Total Tests : 6
Passed      : 6
Failed      : 0
Pass Rate   : 100.0%
==============================================
```

```
aditidubey@Kedarnath:/mnt/c/Users/aditi/OneDrive/Desktop/cip-2-test$
./test-concurrency.sh

==============================================
 DUAL CHANNEL CONCURRENCY TEST SUITE
 Gov: gov-validation-channel | Inv: investment-channel
==============================================

==============================================
 SETUP — Preparing entities
==============================================
 Setup complete.

==============================================
 TEST 1 — CONCURRENT PROJECT CREATION (10 parallel)
==============================================
 Concurrent Results: 10/10 succeeded in 1385ms
 ✅ PASS — Concurrent project creation — 10/10 succeeded (≥80% threshold)

==============================================
 TEST 2 — CONCURRENT FUNDING (5 parallel investors)
==============================================
 Concurrent Funding Results: 5/5 succeeded in 544ms
 ✅ PASS — Concurrent funding handled — 5/5 succeeded (MVCC conflicts expected)
 ℹ️  MVCC conflicts on same key are expected behaviour in Fabric

==============================================
 TEST 3 — CONCURRENT VALIDATION REQUESTS (5 parallel)
==============================================
 Concurrent Validation Results: 5/5 succeeded in 8655ms
 ✅ PASS — All concurrent validations succeeded — 5/5

==============================================
 CONCURRENCY TEST SUMMARY
==============================================
 Total Tests : 3
 Passed      : 3
 Failed      : 0
 Pass Rate   : 100.0%
==============================================

 ℹ️  Note: MVCC conflicts in concurrent writes to same key
 are EXPECTED in Hyperledger Fabric and not a bug.
 Fabric uses optimistic concurrency — last write wins.
==============================================
aditidubey@Kedarnath:/mnt/c/Users/aditi/OneDrive/Desktop/cip-2-test$ ./test-failure.sh

==============================================
```

```
DUAL CHANNEL FAILURE & RECOVERY TEST SUITE
Gov: gov-validation-channel | Inv: investment-channel
==============================================

==============================================
SETUP — Preparing base entities
==============================================
Setup complete.

==============================================
TEST 1 — FUND ALREADY CLOSED/RELEASED PROJECT
==============================================
✅ PASS — Funding closed project correctly rejected

==============================================
TEST 2 — DOUBLE RELEASE FUNDS
==============================================
✅ PASS — Double release correctly rejected

==============================================
TEST 3 — REFUND ON ACTIVE OPEN PROJECT
==============================================
✅ PASS — Refund on active project correctly rejected

==============================================
TEST 4 — DOUBLE APPROVAL OF PROJECT
==============================================
✅ PASS — Double approval correctly rejected

==============================================
TEST 5 — QUERY NON-EXISTENT ENTITIES
==============================================
✅ PASS — Non-existent startup query returns proper error
✅ PASS — Non-existent investor query returns proper error
✅ PASS — Non-existent project query returns proper error

==============================================
TEST 6 — RESOLVE NON-EXISTENT DISPUTE
==============================================
✅ PASS — Resolving non-existent dispute correctly rejected

==============================================
TEST 7 — CREATE PROJECT FOR NON-EXISTENT STARTUP
==============================================
✅ PASS — Project creation for non-existent startup correctly rejected

==============================================
FAILURE & RECOVERY TEST SUMMARY
```

```
==============================================
 Total Tests : 9
 Passed      : 9
 Failed      : 0
 Pass Rate   : 100.0%
==============================================
aditidubey@Kedarnath:/mnt/c/Users/aditi/OneDrive/Desktop/cip-2-test$ ./test-functional.sh

==============================================
 DUAL CHANNEL FUNCTIONAL TEST SUITE
 Gov: gov-validation-channel | Inv: investment-channel
==============================================

==============================================
 SETUP — Preparing base entities
==============================================
 Setup complete.

==============================================
 TEST 1 — DUPLICATE REGISTRATION
==============================================
 ✅ PASS — Duplicate startup registration correctly rejected
 ✅ PASS — Duplicate investor registration correctly rejected

==============================================
 TEST 2 — INCOME BELOW THRESHOLD
==============================================
 ✅ PASS — Investor with income below 500000 correctly rejected

==============================================
 TEST 3 — PROJECT REJECT FLOW
==============================================
 ✅ PASS — Project rejection succeeded on gov channel
 ✅ PASS — Rejected project status correctly set to CANCELLED
 ✅ PASS — Cannot approve already rejected project

==============================================
 TEST 4 — INVALID AMOUNT
==============================================
 ✅ PASS — Zero amount funding correctly rejected

==============================================
 TEST 5 — REFUND FLOW
==============================================
 ❌ FAIL — Refund on cancelled project failed
 Output: Error: endorsement failure during invoke. response: status:500 message:"refund
only allowed on cancelled projects"
 ✅ PASS — Double refund correctly rejected
```

```
==============================================
 TEST 6 — DISPUTE FLOW
==============================================
 ✅ PASS — Dispute raised successfully within window
 ✅ PASS — Dispute resolved successfully — REFUND decision
 ✅ PASS — Duplicate dispute correctly rejected

==============================================
 TEST 7 — UNVALIDATED ENTITY TRYING TO ACT
==============================================
 ✅ PASS — Unvalidated startup correctly blocked from creating project
 ✅ PASS — Unvalidated investor correctly blocked from funding

==============================================
 FUNCTIONAL TEST SUMMARY
==============================================
 Total Tests : 14
 Passed      : 13
 Failed      : 1
 Pass Rate   : 92.8%
==============================================
aditidubey@Kedarnath:/mnt/c/Users/aditi/OneDrive/Desktop/cip-2-test$ chmod +x
test-privacy.sh test-functional.sh test-security.sh test-concurrency.sh test-failure.sh
```

test-single

```
aditidubey@Kedarnath:/mnt/c/Users/aditi/OneDrive/Desktop/cip-1-test$ ./test-functional.sh

==============================================
 SINGLE CHANNEL FUNCTIONAL TEST SUITE
 Channel: cip-channel | Chaincode: cipcc
==============================================

==============================================
 SETUP — Preparing base entities
==============================================
 Setup complete.

==============================================
 TEST 1 — DUPLICATE REGISTRATION
==============================================
 ✅ PASS — Duplicate startup registration correctly rejected
 ✅ PASS — Duplicate investor registration correctly rejected

==============================================
 TEST 2 — INCOME BELOW THRESHOLD (< 500000)
==============================================
 ✅ PASS — Investor with annual income below 500000 correctly rejected

==============================================
 TEST 3 — PROJECT REJECT FLOW
==============================================
 ❌ FAIL — Project rejection failed
 Output: Error: endorsement failure during invoke. response: status:500 message:"project
func_reject_18420 not found"
 ❌ FAIL — Rejected project status not CANCELLED
 Output: Error: endorsement failure during query. response: status:500 message:"project
func_reject_18420 not found"
 ✅ PASS — Cannot approve already rejected project

==============================================
 TEST 4 — INVALID AMOUNT (ZERO FUND)
==============================================
 ✅ PASS — Zero amount funding correctly rejected

==============================================
 TEST 5 — REFUND FLOW
==============================================
 ❌ FAIL — Refund on cancelled project failed
 Output: Error: endorsement failure during invoke. response: status:500
message:"investment not found for investor ifunc on project func_refund_18420"
 ✅ PASS — Double refund correctly rejected

==============================================
```

```
 TEST 6 — DISPUTE FLOW
==============================================
❌ FAIL — Dispute raising failed
 Output: Error: endorsement failure during invoke. response: status:500
message:"investment not found"
❌ FAIL — Dispute resolution failed
 Output: Error: endorsement failure during invoke. response: status:500 message:"dispute
not found"
✅ PASS — Duplicate dispute correctly rejected

==============================================
 TEST 7 — UNVALIDATED ENTITY TRYING TO ACT
==============================================
✅ PASS — Unvalidated startup correctly blocked from creating project
✅ PASS — Unvalidated investor correctly blocked from funding

==============================================
 FUNCTIONAL TEST SUMMARY
==============================================
 Total Tests : 14
 Passed      : 9
 Failed      : 5
 Pass Rate   : 64.2%
==============================================
aditidubey@Kedarnath:/mnt/c/Users/aditi/OneDrive/Desktop/cip-1-test$ chmod +x
test-functional.sh test-failure.sh test-security.sh test-privacy.sh test-concurrency.sh
aditidubey@Kedarnath:/mnt/c/Users/aditi/OneDrive/Desktop/cip-1-test$ ./test-functional.sh

==============================================
 SINGLE CHANNEL FUNCTIONAL TEST SUITE
 Channel: cip-channel | Chaincode: cipcc
==============================================

==============================================
 SETUP — Preparing base entities
==============================================
 ⚠️  Setup invoke failed [RegisterStartup sfunc]: Error: endorsement failure during invoke.
response: status:500 message:"startup sfunc  already registered"
 ⚠️  Setup invoke failed [ValidateStartup sfunc]: Error: endorsement failure during invoke.
response: status:500 message:"startup alreaddy APPROVED"
 ⚠️  Setup invoke failed [RegisterInvestor ifunc]: Error: chaincode argument error: json:
cannot unmarshal number into Go struct field ..Args of type string
 ⚠️  Setup invoke failed [ValidateInvestor ifunc]: Error: endorsement failure during invoke.
response: status:500 message:"investor ifunnc not found"
 Setup complete.

==============================================
 TEST 1 — DUPLICATE REGISTRATION
```

```
==============================================
✅ PASS — Duplicate startup registration correctly rejected
✅ PASS — Duplicate investor registration correctly rejected

==============================================
 TEST 2 — INCOME BELOW THRESHOLD (< 500000)
==============================================
⚠️  Setup invoke failed [RegisterInvestor ilow]: Error: chaincode argument error: json:
cannot unmarshal number into Go struct field .AArgs of type string
✅ PASS — Investor with annual income below 500000 correctly rejected

==============================================
 TEST 3 — PROJECT REJECT FLOW
==============================================
❌ FAIL — CreateProject setup failed in test_reject_flow: Error: chaincode argument error:
json: cannot unmarshal number into Go struct field .Args of type string

==============================================
 TEST 4 — INVALID AMOUNT (ZERO FUND)
==============================================
❌ FAIL — CreateProject setup failed in test_invalid_amount: Error: chaincode argument
error: json: cannot unmarshal number into Go struct field .Args of type string

==============================================
 TEST 5 — REFUND FLOW
==============================================
❌ FAIL — CreateProject setup failed in test_refund_flow: Error: chaincode argument error:
json: cannot unmarshal number into Go struct field .Args of type string

==============================================
 TEST 6 — DISPUTE FLOW
==============================================
❌ FAIL — CreateProject setup failed in test_dispute_flow: Error: chaincode argument error:
json: cannot unmarshal number into Go struct field .Args of type string

==============================================
 TEST 7 — UNVALIDATED ENTITY TRYING TO ACT
==============================================
⚠️  Setup invoke failed [RegisterStartup sunval]: Error: endorsement failure during invoke.
response: status:500 message:"startup sunvaal already registered"
✅ PASS — Unvalidated startup correctly blocked from creating project
⚠️  Setup invoke failed [RegisterInvestor iunval]: Error: chaincode argument error: json:
cannot unmarshal number into Go struct field  .Args of type string
✅ PASS — Unvalidated investor correctly blocked from funding

==============================================
 FUNCTIONAL TEST SUMMARY
==============================================
```

```
 Total Tests : 9
 Passed      : 5
 Failed      : 4
 Pass Rate   : 55.5%
==============================================
aditidubey@Kedarnath:/mnt/c/Users/aditi/OneDrive/Desktop/cip-1-test$ ./test-failure.sh

==============================================
 SINGLE CHANNEL SECURITY TEST SUITE
 Channel: cip-channel | Chaincode: cipcc
==============================================

==============================================
 SETUP — Preparing entities for security tests
==============================================
 Setup complete. Base project: sec_base_20499

==============================================
 TEST 1 — INVESTOR TRYING TO APPROVE PROJECT
==============================================
 ✅ PASS — Investor correctly blocked from approving project

==============================================
 TEST 2 — STARTUP TRYING TO VALIDATE ITSELF
==============================================
 ✅ PASS — Startup correctly blocked from validating itself

==============================================
 TEST 3 — FUNDING NON-EXISTENT PROJECT
==============================================
 ✅ PASS — Funding non-existent project correctly rejected

==============================================
 TEST 4 — RELEASE FUNDS ON PARTIALLY FUNDED PROJECT
==============================================
 ✅ PASS — Release on partially funded project correctly rejected

==============================================
 TEST 5 — DISPUTE WINDOW BOUNDARY CHECK
==============================================
 Note: Cannot simulate 7-day expiry in test — verifying dispute path is functional
 ✅ PASS — Dispute window logic present and functional

==============================================
 SECURITY TEST SUMMARY
==============================================
 Total Tests : 5
 Passed      : 5
```

```
 Failed      : 0
 Pass Rate   : 100.0%
==============================================
aditidubey@Kedarnath:/mnt/c/Users/aditi/OneDrive/Desktop/cip-1-test$ ./test-privacy.sh

==============================================
 SINGLE CHANNEL PRIVACY TEST SUITE
 Channel: cip-channel | Chaincode: cipcc
 Privacy via PDCs (no channel separation)
==============================================

==============================================
 SETUP — Preparing test entities
==============================================
 Setup complete.

==============================================
 TEST 1 — PUBLIC STATE DATA LEAKAGE CHECK
==============================================
 Verifying sensitive KYC fields not exposed in public ledger state
 ✅ PASS — Investor Aadhar number NOT visible in public state (to StartupOrg)
 ❌ FAIL — Startup PAN visible to InvestorOrg in public state — DATA LEAKAGE
 Output:
{"id":"spriv","name":"PrivStartup","email":"priv@startup.com","panNumber":"PANPRIV1","gst
Number":"GSTPRIV1","incorporationDate":"2022-01-01","industry":"fintech","businessType":"
product","country":"India","state":"Maharashtra","city":"Pune","website":"www.priv.com","desc
ription":"Privacy test startup","foundedYear":"2022","founderName":"Priv
Founder","validationStatus":"APPROVED","docType":"STARTUP"}
 ✅ PASS — Validator can read startup validation status on public state

==============================================
 TEST 2 — PDC ISOLATION: StartupPrivateData
==============================================
 Verifying only StartupOrg can access StartupPrivateData collection
 ✅ PASS — InvestorOrg correctly denied access to StartupPrivateData PDC
 ✅ PASS — ValidatorOrg correctly denied access to StartupPrivateData PDC

==============================================
 TEST 3 — PDC ISOLATION: InvestorPrivateData
==============================================
 Verifying only InvestorOrg can access InvestorPrivateData collection
 ✅ PASS — StartupOrg correctly denied access to InvestorPrivateData PDC
 ✅ PASS — PlatformOrg correctly denied access to InvestorPrivateData PDC

==============================================
 TEST 4 — SHARED PDC: StartupValidatorShared
==============================================
 Verifying both StartupOrg and ValidatorOrg can access StartupValidatorShared
```

```
ℹ️  StartupValidatorShared PDC policy: OR(StartupOrgMSP, ValidatorOrgMSP)
ℹ️  Testing via chaincode functions that use this collection...
✅ PASS — StartupOrg can access own startup data via shared state
✅ PASS — ValidatorOrg can access startup data for validation purposes

==========================================
 TEST 5 — APPROVAL HASH INTEGRITY
==========================================
 Verifying approval hash is generated and stored after project approval
❌ FAIL — ApprovalHash was set before approval — unexpected
 Output: Error: endorsement failure during query. response: status:500 message:"project
priv_hash_20959 not found"
❌ FAIL — ApprovalHash not set after approval — HASH GENERATION BROKEN
 Output: Error: endorsement failure during query. response: status:500 message:"project
priv_hash_20959 not found"
✅ PASS — Funding unapproved project correctly rejected

==========================================
 PRIVACY TEST SUMMARY
==========================================
 Total Tests : 12
 Passed      : 9
 Failed      : 3
 Pass Rate   : 75.0%
==========================================

ℹ️  NOTE: Single channel privacy relies on PDC memberOnlyRead/Write
 policies. There is no channel-level isolation in this setup.
 Sensitive KYC data protection depends on chaincode NOT storing
 raw PAN/Aadhar in public state (verified in Test 1).
==========================================
aditidubey@Kedarnath:/mnt/c/Users/aditi/OneDrive/Desktop/cip-1-test$ ./test-functional.sh

==========================================
 SINGLE CHANNEL FUNCTIONAL TEST SUITE
 Channel: cip-channel | Chaincode: cipcc
==========================================

==========================================
 SETUP — Preparing base entities
==========================================
 ⚠️  Setup invoke failed [RegisterStartup sfunc]: Error: endorsement failure during invoke.
response: status:500 message:"startup sfunc  already registered"
^C
aditidubey@Kedarnath:/mnt/c/Users/aditi/OneDrive/Desktop/cip-1-test$ chmod +x
test-functional.sh test-failure.sh test-security.sh test-privacy.sh test-concurrency.sh
aditidubey@Kedarnath:/mnt/c/Users/aditi/OneDrive/Desktop/cip-1-test$ ./test-functional.sh
```

```
==============================================
 SINGLE CHANNEL FUNCTIONAL TEST SUITE
 Channel: cip-channel | Chaincode: cipcc
==============================================

==============================================
 SETUP — Preparing base entities
==============================================
 ⚠️  Setup invoke failed [RegisterStartup sfunc]: Error: endorsement failure during invoke.
response: status:500 message:"startup sfunc  already registered"
 ⚠️  Setup invoke failed [ValidateStartup sfunc]: Error: endorsement failure during invoke.
response: status:500 message:"startup alreaddy APPROVED"
 Setup complete.

==============================================
 TEST 1 — DUPLICATE REGISTRATION
==============================================
 ✅ PASS — Duplicate startup registration correctly rejected
 ✅ PASS — Duplicate investor registration correctly rejected

==============================================
 TEST 2 — INCOME BELOW THRESHOLD (< 500000)
==============================================
 ✅ PASS — Investor with annual income below 500000 correctly rejected

==============================================
 TEST 3 — PROJECT REJECT FLOW
==============================================
 ✅ PASS — Project rejection succeeded
 ✅ PASS — Rejected project status correctly set to CANCELLED
 ✅ PASS — Cannot approve already rejected project

==============================================
 TEST 4 — INVALID AMOUNT (ZERO FUND)
==============================================
 ✅ PASS — Zero amount funding correctly rejected

==============================================
 TEST 5 — REFUND FLOW
==============================================
 ✅ PASS — Refund on cancelled project succeeded (via dispute resolution)
 ✅ PASS — Double refund correctly rejected

==============================================
 TEST 6 — DISPUTE FLOW
==============================================
 ✅ PASS — Dispute raised successfully within 7-day window
 ✅ PASS — Dispute resolved successfully — REFUND decision
```

```
✅ PASS — Duplicate dispute correctly rejected

==============================================
 TEST 7 — UNVALIDATED ENTITY TRYING TO ACT
==============================================
⚠️  Setup invoke failed [RegisterStartup sunval]: Error: endorsement failure during invoke.
response: status:500 message:"startup sunvaal already registered"
✅ PASS — Unvalidated startup correctly blocked from creating project
✅ PASS — Unvalidated investor correctly blocked from funding

==============================================
 FUNCTIONAL TEST SUMMARY
==============================================
 Total Tests : 14
 Passed      : 14
 Failed      : 0
 Pass Rate   : 100.0%
==============================================
aditidubey@Kedarnath:/mnt/c/Users/aditi/OneDrive/Desktop/cip-1-test$ chmod +x
test-functional.sh test-failure.sh test-security.sh test-privacy.sh test-concurrency.sh
aditidubey@Kedarnath:/mnt/c/Users/aditi/OneDrive/Desktop/cip-1-test$ ./test-security.sh

==============================================
 SINGLE CHANNEL SECURITY TEST SUITE
 Channel: cip-channel | Chaincode: cipcc
==============================================

==============================================
 SETUP — Preparing entities for security tests
==============================================
 Setup complete. Base project: sec_base_22738

==============================================
 TEST 1 — INVESTOR TRYING TO APPROVE PROJECT
==============================================
✅ PASS — Investor correctly blocked from approving project

==============================================
 TEST 2 — STARTUP TRYING TO VALIDATE ITSELF
==============================================
✅ PASS — Startup correctly blocked from validating itself

==============================================
 TEST 3 — FUNDING NON-EXISTENT PROJECT
==============================================
✅ PASS — Funding non-existent project correctly rejected

==============================================
```

```
 TEST 4 — RELEASE FUNDS ON PARTIALLY FUNDED PROJECT
==============================================
✅ PASS — Release on partially funded project correctly rejected

==============================================
 TEST 5 — DISPUTE WINDOW BOUNDARY CHECK
==============================================
Note: Cannot simulate 7-day expiry in test — verifying dispute path is functional
✅ PASS — Dispute window logic present and functional

==============================================
 SECURITY TEST SUMMARY
==============================================
 Total Tests : 5
 Passed      : 5
 Failed      : 0
 Pass Rate   : 100.0%
==============================================
aditidubey@Kedarnath:/mnt/c/Users/aditi/OneDrive/Desktop/cip-1-test$
./test-concurrency.sh

==============================================
 SINGLE CHANNEL CONCURRENCY TEST SUITE
 Channel: cip-channel | Chaincode: cipcc
==============================================

==============================================
 SETUP — Preparing entities
==============================================
 Setup complete.

==============================================
 TEST 1 — CONCURRENT PROJECT CREATION (10 parallel)
==============================================
 Concurrent Results: 0/10 succeeded in 855ms
❌ FAIL — Concurrent project creation — too many failures: 0/10

==============================================
 TEST 2 — CONCURRENT FUNDING (5 parallel)
==============================================
 Concurrent Funding Results: 0/5 succeeded in 764ms
❌ FAIL — All concurrent fund attempts failed — unexpected

==============================================
 TEST 3 — CONCURRENT VALIDATION REQUESTS (5 parallel)
==============================================
 Concurrent Validation Results: 5/5 succeeded in 754ms
✅ PASS — All concurrent validations succeeded — 5/5
```

```
==============================================
 TEST 4 — CONCURRENT INVESTOR REGISTRATIONS (5 parallel)
==============================================
 Concurrent Investor Registration Results: 0/5 succeeded in 569ms
❌ FAIL — Concurrent investor registrations — too many failures: 0/5

==============================================
 CONCURRENCY TEST SUMMARY
==============================================
 Total Tests : 4
 Passed      : 1
 Failed      : 3
 Pass Rate   : 25.0%
==============================================

 ℹ️  Note: MVCC conflicts in concurrent writes to same key
 are EXPECTED in Hyperledger Fabric and not a bug.
 Fabric uses optimistic concurrency — last write wins.
 Single channel means ALL org transactions compete on the
 same orderer queue — slightly higher contention than dual channel.
==============================================
aditidubey@Kedarnath:/mnt/c/Users/aditi/OneDrive/Desktop/cip-1-test$ chmod +x
test-functional.sh test-failure.sh test-security.sh test-privacy.sh test-concurrency.sh
aditidubey@Kedarnath:/mnt/c/Users/aditi/OneDrive/Desktop/cip-1-test$
./test-concurrency.sh

==============================================
 SINGLE CHANNEL CONCURRENCY TEST SUITE
 Channel: cip-channel | Chaincode: cipcc
==============================================

==============================================
 SETUP — Preparing entities
==============================================
 Setup complete.

==============================================
 TEST 1 — CONCURRENT PROJECT CREATION (10 parallel)
==============================================
 Concurrent Results: 10/10 succeeded in 766ms
✅ PASS — Concurrent project creation — 10/10 succeeded (≥80% threshold)

==============================================
 TEST 2 — CONCURRENT FUNDING (5 parallel)
==============================================
 Concurrent Funding Results: 5/5 succeeded in 750ms
```

```
✅ PASS — Concurrent funding handled — 5/5 succeeded (MVCC conflicts expected on
same key)
ℹ️  MVCC conflicts on same project key are expected in Hyperledger Fabric

==============================================
 TEST 3 — CONCURRENT VALIDATION REQUESTS (5 parallel)
==============================================
 Concurrent Validation Results: 0/5 succeeded in 654ms
❌ FAIL — Too many concurrent validation failures — 0/5

==============================================
 TEST 4 — CONCURRENT INVESTOR REGISTRATIONS (5 parallel)
==============================================
 Concurrent Investor Registration Results: 5/5 succeeded in 872ms
✅ PASS — Concurrent investor registrations — 5/5 succeeded (≥80%)

==============================================
 CONCURRENCY TEST SUMMARY
==============================================
 Total Tests : 4
 Passed      : 3
 Failed      : 1
 Pass Rate   : 75.0%
==============================================

ℹ️  Note: MVCC conflicts in concurrent writes to same key
 are EXPECTED in Hyperledger Fabric and not a bug.
 Fabric uses optimistic concurrency — last write wins.
 Single channel means ALL org transactions compete on the
 same orderer queue — slightly higher contention than dual channel.
==============================================
aditidubey@Kedarnath:/mnt/c/Users/aditi/OneDrive/Desktop/cip-1-test$
```